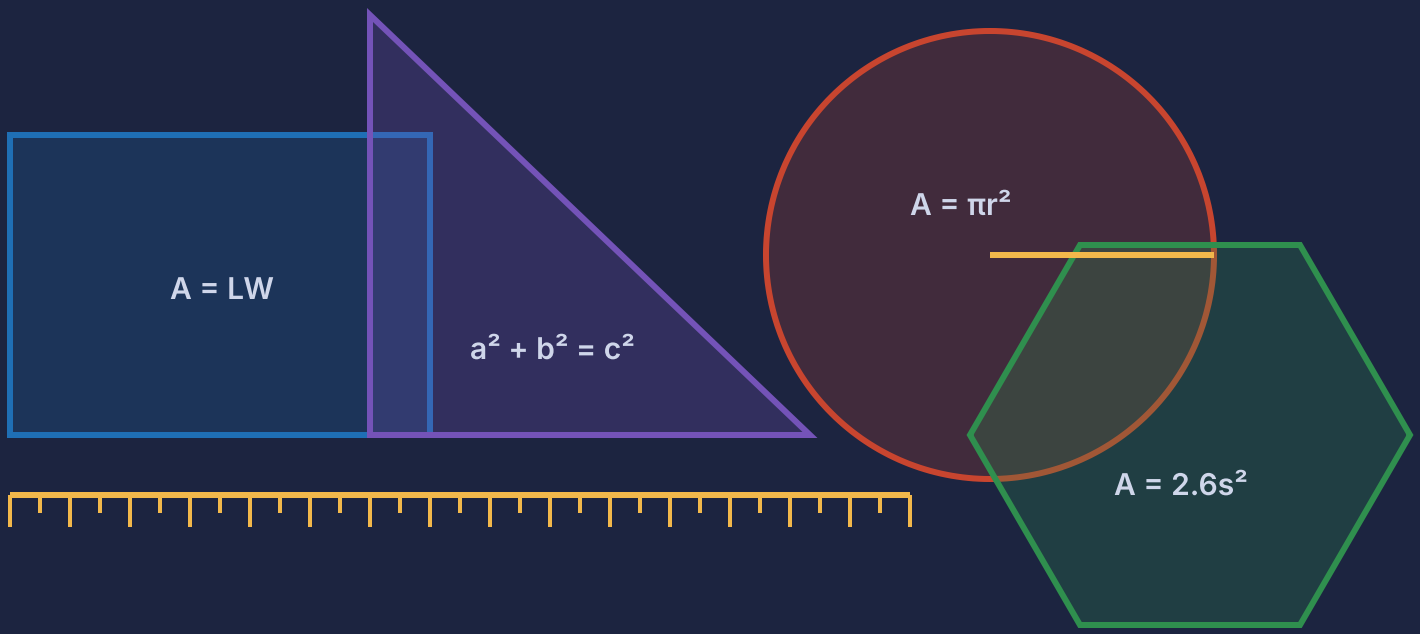
A FULL-COLOR COURSE TEXT

Applied Technical Math

Numbers, Measurement, Algebra and Geometry for the Trades



NUMBERS · MEASUREMENT · ALGEBRA · GEOMETRY

FALL 2026

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How to use this book

This book follows one semester of applied technical math, in the order the course is taught: numbers first, then measurement, then algebra with formulas, then the geometry of flat shapes. Every topic is taught the same way.

KEY RULE **Blue boxes**

The rule or formula to learn. If a page is read for one thing only, read these.

WATCH OUT **Red boxes**

The mistakes that cost the most points, and how to avoid each one.

CALCULATOR TIP **Purple boxes**

Which keys to press, and what the calculator will not do for you.

ON THE JOB **Green boxes**

Where the skill shows up in the trades: sheet metal, piping, electrical, machining, building.

Worked examples show the work on the left and the reason for each line on the right. Cover the right-hand column and explain each line yourself; that is the fastest way to find out what you do not yet understand.

Practice sets follow each section and a **chapter review** closes each chapter. Answers to every problem are at the back of the book.

Conventions used in the course

- **Show the substitution.** When a formula is used, write the formula, then write it again with the numbers in place of the letters, then calculate. Every example in this book does this.
- **Round only at the end,** and only as far as the problem says. If a problem does not say, do not round.
- **Use 3.14 for π** unless a problem says otherwise.
- **Write the unit.** A number without a unit is not a measurement: 12 could be inches, feet, or volts.
- **Fractions are given in lowest terms,** and a mixed number is preferred to an improper fraction ($1\frac{3}{8}$ in rather than $\frac{11}{8}$ in).

APPLIED TECHNICAL MATH

Part 1 · Numbers

Every measurement, price and specification is a number written in the base ten system. Part 1 makes sure those numbers are read, rounded and rewritten correctly before any calculating begins.

CHAPTER 1

Place Value and Rounding

A micrometer reads 0.3175 inch. A blueprint calls for 0.318. Are they the same? The answer depends on knowing what each digit is worth and how rounding works. This chapter is the foundation for everything else in the book.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Name the place value of any digit, from millions down to ten-thousandths.
- Read and write decimal numbers the way they are spoken in a shop.
- Compare two decimals and say which is larger.
- Round any number to a named place using the five-step rule.
- Decide how far to round from the wording of a problem.

KEY TERMS

digit	One of the ten symbols 0 to 9.
place value	What a digit is worth because of where it sits in the number.
decimal point	The dot that separates the whole-number part from the part smaller than one.
rounding digit	The digit in the place you are rounding to.
test digit	The digit just to the right of the rounding digit. It decides whether to round up.

1.1 The base ten system

Our number system uses only ten digits, yet it can write any amount. It does this by giving each *position* in a number its own value. Moving one place to the left makes a digit worth **ten times as much**. Moving one place to the right makes it worth **one tenth as much**. That single idea runs straight through the decimal point without changing.

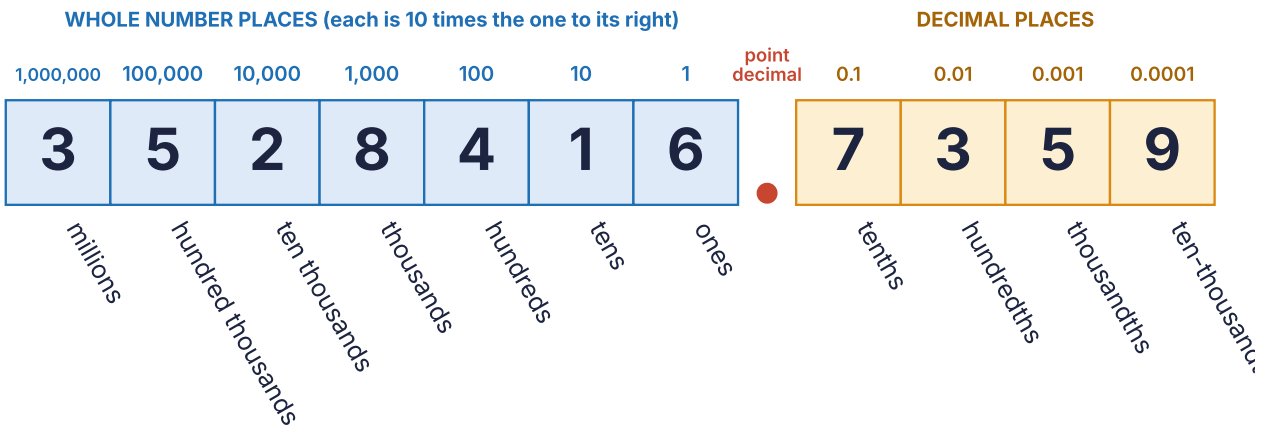


Figure 1.1 The place values used in this course, shown with the number 3,528,416.7359.

In the number above, the digit 5 sits in the hundred thousands place, so it is worth 500,000. Another 5 sits in the thousandths place and is worth only 0.005. Same digit, very different value.

KEY RULE **Whole-number places end in s; decimal places end in ths**

To the **left** of the decimal point: ones, tens, hundreds, thousands, ten thousands, hundred thousands, millions.

To the **right** of the decimal point: tenths, hundredths, thousandths, ten-thousandths.

There is no "oneths" place. The first place to the right of the point is tenths.

EXAMPLE 1.1

In the number 902,468.7351, name the digit in the (a) ten thousands place, (b) hundredths place, (c) ten-thousandths place.

902,468 . 7351	Find the decimal point first. Every place is counted from it.
9 0 2, 4 6 8 → ones 8, tens 6, hundreds 4, thousands 2, ten thousands **0**	Count left from the point: ones, tens, hundreds, thousands, ten thousands.
7 3 5 1 → tenths 7, hundredths **3**	Count right from the point: tenths, hundredths.
thousandths 5, ten-thousandths **1**	Keep counting right for the last two places.

ANSWER **(a) 0 (b) 3 (c) 1**

WATCH OUT **A zero is still a digit**

In 902,468.7351 the ten thousands digit is 0. A zero holds its place so that the 9 stays in the hundred thousands place. Never skip over a zero when counting places.

1.2 Reading and writing decimals

A decimal is read in three parts: the whole number, the word **and** for the decimal point, then the decimal part read as if it were a whole number followed by the name of its **last** place.

Number	Read as
0.7	seven tenths
0.25	twenty-five hundredths
3.125	three and one hundred twenty-five thousandths
12.06	twelve and six hundredths
0.0625	six hundred twenty-five ten-thousandths

ON THE JOB **How decimals are spoken in a machine shop**

Machinists measure in thousandths of an inch, so they read almost everything in thousandths. The dimension 0.250 in is “two hundred fifty thousandths,” and 0.0005 in is “five tenths” (short for five tenths of a thousandth). The value has not changed, only the way it is said.

Zeros written at the right-hand end of a decimal do not change its value: 0.5, 0.50 and 0.500 are the same amount. On a drawing, however, the extra zeros tell the reader how precisely the part must be made, so copy them exactly.

1.3 Comparing decimals

To decide which of two decimals is larger, line up the decimal points and compare digit by digit from the left. The first place where the digits differ settles it. Filling in zeros so that both numbers have the same number of decimal places makes the comparison easy to see.

EXAMPLE 1.2

Which is larger, 0.8125 or 0.813?

0.8125 0.8130	Write both with four decimal places. Adding a zero to 0.813 does not change it.
0.81**2**5 0.81**3**0	The tenths (8) and hundredths (1) match. The thousandths differ: 2 against 3.
3 > 2	The number with the larger digit in that place is the larger number.

ANSWER **0.813 is larger.**

WATCH OUT **More digits does not mean a bigger number**

0.8125 has more digits than 0.813 but it is the smaller number. Length only matters for whole numbers.

PRACTICE 1.1 **Place value**

For problems 1 to 10 use the number 4,107,538.2069. Name the digit in each place. For 11 to 16, tell which number is larger.

1. the thousands place
2. the tenths place
3. the ten-thousandths place
4. the millions place
5. the hundredths place
6. the ten thousands place
7. the ones place
8. the thousandths place
9. the hundred thousands place
10. the tens place
11. 0.5 or 0.45
12. 0.062 or 0.0625
13. 1.3 or 1.29
14. 0.4375 or 0.44
15. 2.005 or 2.05
16. 0.75 or 0.705

1.4 Rounding

Rounding replaces a number with a nearby number that is easier to use. A tape measure cannot show 37.4286 inches, and a supplier will not sell 71,428.57 psi steel. The question is always the same: **which round value is the number closest to?**



Figure 1.2 Rounding 9372.6 to the nearest ten. It lies between 9370 and 9380, and it is closer to 9370.

KEY RULE **The five steps of rounding**

1. Find the **rounding digit**, the digit in the place you are rounding to. Underline it.

2. Look at the **test digit**, the one digit immediately to its right. Ignore everything after that.

3. If the test digit is **5, 6, 7, 8 or 9**, round up: add one to the rounding digit.

4. If the test digit is **0, 1, 2, 3 or 4**, leave the rounding digit as it is.

5. Replace what follows. In a whole number, digits after the rounding digit become **zeros**. After the decimal point, digits after the rounding digit are **dropped**.

EXAMPLE 1.3

Round 9372.6485 to the nearest (a) ten, (b) hundredth, (c) thousand.

93**7**2.6485 → test digit 2	(a) The tens digit is 7. The digit to its right is 2, which is 4 or less, so the 7 stays.
9370	The ones digit becomes a zero and the decimal part is dropped.
9372.6**4**85 → test digit 8	(b) The hundredths digit is 4. The test digit 8 is 5 or more, so the 4 becomes 5.
9372.65	Everything after the hundredths place is dropped.
9372.6485 → test digit 3	(c) The thousands digit is 9. The test digit 3 leaves it alone.
9000	The three whole-number places after it become zeros.

ANSWER **(a) 9370 (b) 9372.65 (c) 9000**

EXAMPLE 1.4 When the rounding digit is 9

Round 4.3972 to the nearest hundredth.

4.39**7**2 → test digit 7	The hundredths digit is 9 and the test digit says round up.
9 + 1 = 10	Adding one to 9 makes ten, so write 0 and carry 1 to the tenths place.
4.40	The tenths digit 3 becomes 4. Keep the zero: it shows the number was rounded to hundredths.

ANSWER **4.40**

WATCH OUT **Round once, from the original number**

To round 2.3491 to the nearest tenth, the test digit is 4, so the answer is 2.3. Do not round in stages (2.3491 → 2.35 → 2.4). Rounding twice gives the wrong answer.

The names used for rounding places vary. All of these mean the same thing:

The problem says	Round to this place	Decimal places kept
nearest whole number, nearest unit, nearest one	ones	0
nearest tenth, nearest 10th, one decimal place	tenths	1
nearest hundredth, nearest 100th, two decimal places	hundredths	2
nearest thousandth, three decimal places	thousandths	3
nearest ten	tens	none; ends in 0
nearest hundred	hundreds	none; ends in 00

WATCH OUT **Ten or tenth? Hundred or hundredth?**

Read the last three letters. "Nearest **hundred**" rounds 3925 to 3900. "Nearest **hundredth**" rounds 3.925 to 3.93. Mixing these up is one of the most common ways to lose points on a correct calculation.

CALCULATOR TIP **Let the calculator do the arithmetic, not the rounding**

Write down more digits than you need from the display, then round by hand using the five steps. Keep the full number in the calculator if the result is needed for another step: rounding in the middle of a problem can change the final answer.

PRACTICE 1.2 **Rounding**

Round each number to the place named.

1. 638.2749 to the nearest tenth
2. 638.2749 to the nearest hundredth
3. 638.2749 to the nearest ten
4. 638.2749 to the nearest hundred
5. 15,862.0953 to the nearest thousand
6. 15,862.0953 to the nearest thousandth
7. 15,862.0953 to the nearest whole number
8. 15,862.0953 to the nearest tenth
9. 0.9968 to the nearest hundredth
10. 0.9968 to the nearest tenth
11. 7,449.5 to the nearest whole number
12. 7,449.5 to the nearest hundred
13. 86.0451 to the nearest hundredth
14. 249.96 to the nearest tenth
15. 3,925 to the nearest hundred
16. 71,428.57 to the nearest hundred
17. 55.901 to the nearest whole number
18. 0.04449 to the nearest thousandth

1.5 How far should an answer be rounded?

In this course the problem always says. Look for the instruction in parentheses or at the top of the page, and follow it exactly. When there is no instruction, give the full answer.

- **"Round to the nearest tenth if needed"** means: if the answer comes out even (139.5) or whole (3456), leave it alone. Round only if there are more digits than asked for.
- **"Give a non-rounded decimal or a fraction in lowest terms"** means rounding is not allowed. Write every digit, or use a fraction.
- **"Nearest whole number"** in a counting problem (how many pipes, how many sheets) may really mean *round up*, because 15.6 pipes cannot be bought. The problem will say if it wants that.

ON THE JOB **Rounding costs money**

A concrete order of 92.59 cubic yards rounded *down* to 92 leaves the pour short. Rounded to the nearest tenth it is 92.6. On the job, know whether being a little over or a little under is the safer mistake.

CHAPTER SUMMARY

- Each place is worth ten times the place to its right. Places left of the point end in **s**; places right of it end in **ths**.
- To compare decimals, line up the points and compare from the left.
- To round: find the rounding digit, look only at the next digit, 5 or more rounds up, 4 or less stays.
- After rounding, whole-number places fill with zeros and decimal places are dropped.
- Round once, at the end, and only as far as the problem says.

CHAPTER 1 REVIEW

1. In 6,052,917.3084, which digit is in the hundredths place?
2. In 6,052,917.3084, which digit is in the ten thousands place?
3. In 6,052,917.3084, which digit is in the thousandths place?
4. In 6,052,917.3084, which digit is in the hundred thousands place?
5. In 6,052,917.3084, which digit is in the ten-thousandths place?
6. In 6,052,917.3084, which digit is in the tenths place?
7. Round 5,274.9638 to the nearest tenth.
8. Round 5,274.9638 to the nearest hundred.
9. Round 5,274.9638 to the nearest thousandth.
10. Round 5,274.9638 to the nearest whole number.
11. Round 5,274.9638 to the nearest ten.
12. Round 5,274.9638 to the nearest hundredth.
13. Which is larger, 0.3125 or 0.32?
14. Which is smaller, 1.07 or 1.068?
15. Write in digits: forty-two and seven hundredths.
16. Write in digits: three hundred seventy-five thousandths.
17. A calculation gives 2,866.2 rpm. Round to the nearest hundred rpm.
18. A calculation gives 7.4667 microfarads. Round to the nearest hundredth.
19. A student rounds 6.2481 to the nearest tenth and writes 6.3. What went wrong, and what is the answer?
20. Round 19.996 to the nearest hundredth.

CHAPTER 2

Fractions, Decimals and Percents

Five-eighths of an inch, 0.625 inch and 62.5% of an inch are the same length written three ways. Tape measures use fractions, calculators use decimals, and specifications use percents. A technician has to translate among them without hesitation.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Reduce a fraction to lowest terms and change between improper fractions and mixed numbers.
- Change a fraction to a decimal and a decimal to a fraction.
- Change a decimal to a percent and a percent to a decimal.
- Move freely among all three forms and recognise the common shop equivalents.
- Add and subtract measurements that mix fractions and decimals.

KEY TERMS

numerator	The top number of a fraction: how many parts are counted.
denominator	The bottom number: how many equal parts make one whole.
lowest terms	A fraction whose top and bottom share no common factor except 1.
mixed number	A whole number and a fraction together, such as $2\ \frac{3}{8}$.
improper fraction	A fraction whose top is as large as or larger than its bottom, such as $\frac{19}{8}$.
percent	Parts out of one hundred. 25% means 25 out of every 100.

2.1 Fractions in lowest terms

A fraction names part of a whole. The **denominator** says how many equal parts the whole was cut into, and the **numerator** says how many of those parts are being counted. On an inch ruler cut into sixteenths, the mark $\frac{12}{16}$ is the same mark as $\frac{6}{8}$ and $\frac{3}{4}$. All three are **equivalent fractions**, but only $\frac{3}{4}$ is in lowest terms.

KEY RULE **Reducing to lowest terms**

Divide the top and the bottom by the same number, and keep going until no number divides both.

Ruler fractions have denominators 2, 4, 8, 16, 32 and 64, so for them the rule is short: **keep halving top and bottom until the top is odd.**

EXAMPLE 2.1

Reduce $\frac{20}{32}$ to lowest terms.

$\frac{20}{32} = \frac{20 \div 2}{32 \div 2} = \frac{10}{16}$	Both are even, so divide both by 2.
$\frac{10}{16} = \frac{5}{8}$	Still both even. Divide by 2 again.
$\frac{5}{8}$	The top is odd, so the fraction is in lowest terms.

ANSWER $\frac{5}{8}$

Mixed numbers and improper fractions

A measurement such as $2\ \frac{3}{8}$ inches is a **mixed number**. For multiplying and dividing it is easier to use the **improper fraction** $\frac{19}{8}$. Both say the same thing: nineteen eighths.

> Mixed → improper: $2\ \frac{3}{8} = \frac{2 \times 8 + 3}{8} = \frac{19}{8}$ (whole × bottom + top, over the same bottom)
>
> Improper → mixed: $\frac{19}{8} = 19 \div 8 = 2$ remainder $3 = 2\ \frac{3}{8}$ (divide; the remainder goes on top)

PRACTICE 2.1 **Fractions**

Reduce 1 to 8 to lowest terms. Write 9 to 12 as improper fractions and 13 to 16 as mixed numbers.

1. $\frac{6}{8}$
2. $\frac{12}{16}$
3. $\frac{10}{32}$
4. $\frac{24}{64}$
5. $\frac{14}{16}$
6. $\frac{28}{32}$
7. $\frac{36}{64}$
8. $\frac{4}{16}$

9. $1\frac{3}{4}$
10. $2\frac{5}{8}$
11. $3\frac{7}{16}$
12. $5\frac{1}{2}$
13. $\frac{11}{8}$
14. $\frac{9}{4}$
15. $\frac{37}{16}$
16. $\frac{45}{32}$

2.2 Fraction to decimal

The fraction bar is a division sign. To change any fraction to a decimal, do the division it shows.

KEY RULE **Fraction → decimal**

Divide the top by the bottom.

$\frac{5}{8} = 5 \div 8 = 0.625$

For a mixed number, keep the whole number and convert only the fraction: $3\frac{5}{8} = 3 + 0.625 = 3.625$. Some divisions never end. $\frac{1}{3} = 0.3333\ldots$ and $\frac{1}{6} = 0.1666\ldots$ For those, round as the problem directs.

EXAMPLE 2.2

Change 31 $\frac{3}{8}$ in to a decimal.

3 ÷ 8 = 0.375	Divide the top of the fraction by the bottom.
31 + 0.375 = 31.375	Put the whole number back in front.

ANSWER **31.375 in**

CALCULATOR TIP **Entering a mixed number**

On any calculator, $31\frac{3}{8}$ can be keyed as `31 + 3 ÷ 8`. The calculator divides before it adds, so the answer is 31.375. Inside a longer calculation, put it in parentheses: `(31 + 3 ÷ 8)`.

2.3 Decimal to fraction

Read the decimal by its place value and write exactly what you say. Then reduce.

KEY RULE **Decimal → fraction**

Write the digits after the point over 10, 100, 1000 or 10,000: a 1 followed by one zero for each decimal place. Then reduce.

0.375 = "375 thousandths" = $\frac{375}{1000} = \frac{3}{8}$

EXAMPLE 2.3

Write 0.4375 as a fraction in lowest terms.

$0.4375 = \frac{4375}{10{,}000}$	Four decimal places, so the bottom is 1 with four zeros.
$\frac{4375 \div 625}{10{,}000 \div 625} = \frac{7}{16}$	Divide top and bottom by 5 four times (5 × 5 × 5 × 5 = 625).
7 ÷ 16 = 0.4375 ✓	Check by dividing.

ANSWER $\frac{7}{16}$

CALCULATOR TIP **The fraction key**

Most scientific calculators have a key that converts a decimal answer to a fraction (often marked `a b/c`, `F↔D` or `▸Frac`). It is the fastest way to reduce, but know the hand method too: the key cannot help when the decimal has been rounded.

Decimal to the nearest ruler fraction

A calculated length such as 2.70 in is not a mark on a tape. To find the nearest 16th, multiply the decimal part by 16 and round to a whole number.

EXAMPLE 2.4

What is 2.70 in to the nearest 16th of an inch?

0.70 × 16 = 11.2	Multiply only the decimal part by the denominator wanted.
11.2 → 11	Round to the nearest whole number. That is the number of sixteenths.
$2\ \frac{11}{16}$ in	Put the whole inches back. Reduce if the top came out even.

ANSWER $\mathbf{2\ \frac{11}{16}}$ **in**

2.4 Decimals and percents

Percent means "per hundred." The symbol % stands for "÷ 100," so 37.5% is 37.5 ÷ 100 = 0.375. Because dividing or multiplying by 100 only shifts the decimal point two places, the conversions can be done by sight.

KEY RULE **Decimal ↔ percent**

Decimal → percent: move the decimal point **2 places right** and add the % sign (the same as × 100). 0.167 = 16.7%

Percent → decimal: move the decimal point **2 places left** and drop the % sign (the same as ÷ 100). 37.5% = 0.375

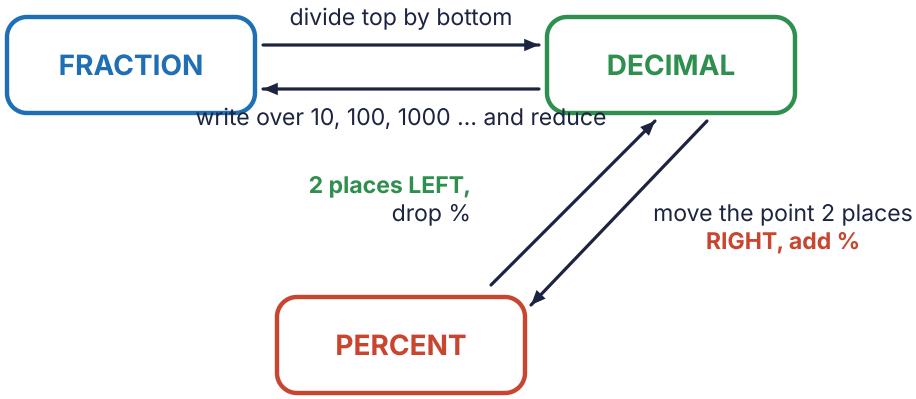


Figure 2.1 The three forms and how to move between them. To go from a fraction to a percent, go through the decimal.

EXAMPLE 2.5

Write each as a percent: (a) 0.08 (b) 1.25 (c) 0.004

0.08 → 0 0 8. → 8%	(a) Two places right. The point lands after the 8.
1.25 → 1 2 5. → 125%	(b) A decimal larger than 1 is a percent larger than 100.
0.004 → 0 0 0.4 → 0.4%	(c) Two places only. This is less than one percent.

ANSWER **(a) 8% (b) 125% (c) 0.4%**

WATCH OUT **Which way does the point move?**

A percent is always the bigger-looking number: 0.25 is 25%. If you are going **to** percent the number should look larger, so the point moves right. Going **from** percent it should look smaller, so the point moves left.

A single-digit percent needs a zero filled in: 6% = 0.06, not 0.6.

2.5 All three forms together

Any one form leads to the other two. The path always passes through the decimal.

EXAMPLE 2.6

Complete the row: fraction $\frac{5}{8}$ → decimal → percent.

$5 \div 8 = 0.625$	Top divided by bottom.
$0.625 \rightarrow 62.5\%$	Point two places right, add %.

ANSWER $\frac{5}{8} = 0.625 = 62.5\%$

EXAMPLE 2.7

Complete the row: 37.5% → decimal → fraction.

$37.5\% \rightarrow 0.375$	Point two places left, drop %.
$0.375 = \frac{375}{1000}$	Three decimal places: over 1000.
$\frac{375 \div 125}{1000 \div 125} = \frac{3}{8}$	Reduce.

ANSWER $37.5\% = 0.375 = \frac{3}{8}$

Table 2.1 Equivalents worth knowing by heart (thirds rounded to four places)

Fraction	Decimal	Percent	Fraction	Decimal	Percent
$\frac{1}{16}$	0.0625	6.25%	$\frac{1}{8}$	0.125	12.5%
$\frac{3}{16}$	0.1875	18.75%	$\frac{1}{4}$	0.25	25%
$\frac{5}{16}$	0.3125	31.25%	$\frac{3}{8}$	0.375	37.5%
$\frac{7}{16}$	0.4375	43.75%	$\frac{1}{2}$	0.5	50%
$\frac{9}{16}$	0.5625	56.25%	$\frac{5}{8}$	0.625	62.5%
$\frac{11}{16}$	0.6875	68.75%	$\frac{3}{4}$	0.75	75%
$\frac{13}{16}$	0.8125	81.25%	$\frac{7}{8}$	0.875	87.5%
$\frac{15}{16}$	0.9375	93.75%	$\frac{1}{3}$	0.3333	33.33%
$\frac{2}{3}$	0.6667	66.67%	$\frac{1}{5}$	0.2	20%

PRACTICE 2.2

Changing forms

Write each number in the other two forms. Give fractions in lowest terms and do not round.

1. $\frac{3}{8}$ (to a decimal and a percent)
2. 0.8 (to a fraction and a percent)
3. 45% (to a decimal and a fraction)
4. $\frac{7}{16}$ (to a decimal and a percent)
5. 0.065 (to a fraction and a percent)
6. 12.5% (to a decimal and a fraction)
7. $\frac{9}{20}$ (to a decimal and a percent)
8. 1.5 (to a fraction and a percent)
9. 2% (to a decimal and a fraction)
10. $\frac{13}{16}$ (to a decimal and a percent)
11. 0.3125 (to a fraction and a percent)
12. 87.5% (to a decimal and a fraction)

2.6 Adding and subtracting mixed measurements

Drawings often give some dimensions as fractions and others as decimals. Before adding or subtracting, put every number in the same form. Decimals are usually quicker; fractions are exact and match the tape.

Method 1: change everything to decimals

EXAMPLE 2.8

A bracket is 11.25 in long. A second piece 31 $\frac{3}{8}$ in long is welded to it. What is the total length?

$31\ \frac{3}{8} = 31.375$	Change the mixed number to a decimal: 3 ÷ 8 = 0.375.
$11.250 + 31.375 = 42.625$	Line up the decimal points and add.
$0.625 = \frac{5}{8}$	If a fraction is wanted, change the decimal part back.

ANSWER **42.625 in, or 42 $\frac{5}{8}$ in**

Method 2: keep everything in fractions

Fractions can be added or subtracted only when their denominators match. For ruler fractions the common denominator is simply the **larger** of the two, because each denominator is a multiple of the smaller ones.

EXAMPLE 2.9

$\frac{7}{8} - \frac{3}{16}$

$\frac{7}{8} = \frac{7 \times 2}{8 \times 2} = \frac{14}{16}$	Change eighths to sixteenths: multiply top and bottom by 2.
$\frac{14}{16} - \frac{3}{16} = \frac{11}{16}$	Same denominator, so subtract the tops. The bottom does not change.

ANSWER $\frac{11}{16}$

EXAMPLE 2.10 Borrowing

$5\frac{1}{4} - 2\frac{5}{8}$

$5\frac{2}{8} - 2\frac{5}{8}$	Common denominator 8.
$4\frac{10}{8} - 2\frac{5}{8}$	2 is smaller than 5, so borrow 1 from the 5. One whole is $\frac{8}{8}$, and $\frac{8}{8} + \frac{2}{8} = \frac{10}{8}$.
$2\frac{5}{8}$	Subtract the whole numbers (4 − 2) and the fractions ($\frac{10}{8} - \frac{5}{8}$).

ANSWER $2\frac{5}{8}$

WATCH OUT **Never add the denominators**

$\frac{1}{4} + \frac{1}{4}$ is $\frac{2}{4} = \frac{1}{2}$, not $\frac{2}{8}$. The denominator names the size of the pieces, and the size does not change when pieces are added.

PRACTICE 2.3 **Adding and subtracting**

Give fraction answers in lowest terms as mixed numbers.

1. $\frac{3}{8} + \frac{1}{4}$
2. $\frac{5}{16} + \frac{1}{2}$
3. $\frac{7}{8} - \frac{3}{16}$
4. $\frac{3}{4} - \frac{5}{16}$
5. $2\frac{1}{2} + 1\frac{3}{8}$
6. $4\frac{3}{16} + 2\frac{7}{8}$
7. $6\frac{1}{4} - 2\frac{5}{8}$
8. $10 - 3\frac{7}{16}$
9. 14.5 in + $6\frac{3}{4}$ in (decimal answer)
10. $22\frac{5}{8}$ in + 9.25 in (decimal answer)
11. $48\frac{3}{4}$ in − 17.125 in (decimal answer)
12. 60.5 in − $24\frac{7}{8}$ in (decimal answer)

CHAPTER SUMMARY

- A fraction is a division: top ÷ bottom gives the decimal.
- A decimal becomes a fraction by writing it over 10, 100, 1000 … and reducing.
- Decimal to percent: point 2 places right, add %. Percent to decimal: point 2 places left, drop %.
- To go between a fraction and a percent, pass through the decimal.
- Add or subtract fractions only after giving them the same denominator; never add denominators.

CHAPTER 2 REVIEW

1. Reduce $\frac{18}{32}$ to lowest terms.
2. Reduce $\frac{40}{64}$ to lowest terms.
3. Change $\frac{7}{8}$ to a decimal.
4. Change $\frac{3}{16}$ to a decimal.
5. Change $2\ \frac{9}{16}$ to a decimal.
6. Change 0.625 to a fraction in lowest terms.
7. Change 0.35 to a fraction in lowest terms.
8. Change 0.1875 to a fraction in lowest terms.
9. Change 0.42 to a percent.
10. Change 0.075 to a percent.
11. Change 1.8 to a percent.
12. Change 64% to a decimal.
13. Change 7% to a decimal.
14. Change 112.5% to a decimal.
15. Change 0.5% to a decimal.
16. Change $\frac{11}{16}$ to a percent.
17. Change 62.5% to a fraction in lowest terms.
18. What is 5.40 in to the nearest 16th of an inch?
19. $3\ \frac{5}{8} + 2\ \frac{9}{16}$
20. $8\ \frac{1}{8} - 3\ \frac{3}{4}$
21. A shaft 18.375 in long has $\frac{7}{16}$ in cut from one end. What length is left, as a decimal?
22. Three spacers measure $\frac{3}{8}$ in, 0.25 in and $\frac{9}{16}$ in. What is their total thickness as a fraction?
23. Which is larger, $\frac{13}{16}$ or 0.81?

CHAPTER 3

The Percent Proportion

Every percent problem is the same sentence with one number missing: *the part is some percent of the whole*. Learn to find the three pieces in that sentence, and one proportion solves them all.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Identify the part, the whole and the percent in a percent statement.
- Set up the percent proportion and solve it by cross-multiplying.
- Solve all three types of percent problem with one method.
- Use the decimal shortcut for each type.
- Solve applied problems: efficiency, discount, tax, waste and percent change.

KEY TERMS

part	The piece being compared. It goes with the word **is**.
whole	The total or original amount the part is compared to. It follows the word **of**. Also called the base.
percent	The number with the % sign. It always sits over 100 in the proportion.
proportion	A statement that two fractions are equal.
cross-multiply	Multiply the two numbers that sit diagonally across the equals sign.

3.1 Part, whole and percent

Read this sentence slowly: **45 is 25% of 180.** It has three numbers, and each has a job.

- **180** is the **whole**. It comes right after the word *of*. It is the total, the starting amount, the 100%.
- **45** is the **part**. It goes with the word *is*. It is the piece of the whole being talked about.
- **25%** is the **percent**. It is the easiest to find because it wears the % sign.



Figure 3.1 45 is 25% of 180. The part fills 25 hundredths of the whole.

KEY RULE **The percent proportion**

$$\frac{\text{PART (is)}}{\text{WHOLE (of)}} = \frac{\text{PERCENT}}{100}$$

The part compared to the whole is the same comparison as the percent compared to 100.

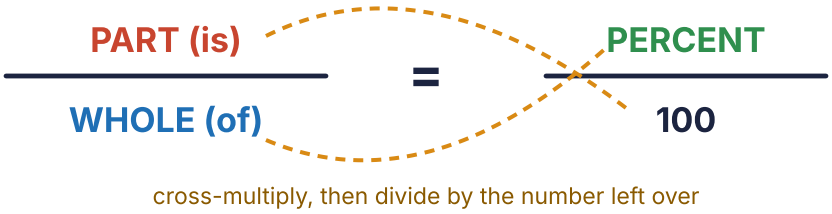


Figure 3.2 Is over of equals percent over 100.

In a problem, one of the three is unknown. Call it x, put the two known numbers in their places, and solve.

WATCH OUT **The whole is not always the bigger number**

In "150 is 125% of what number?" the part (150) is larger than the whole (120). Do not decide by size. Decide by the words: the whole follows **of**, the part goes with **is**.

3.2 Solving the proportion

KEY RULE **Cross-multiply and divide**

1. Multiply the two numbers that are diagonal from each other.

2. Divide by the number that is left over (the one diagonal from x).

This works wherever x happens to be.

Type 1: finding the part

EXAMPLE 3.1

What is 25% of 180?

$\frac{x}{180} = \frac{25}{100}$	180 follows *of*, so it is the whole. 25 has the % sign. The part is unknown.
$25 \times 180 = 4500$	Cross-multiply the diagonal pair that has two numbers.
$4500 \div 100 = 45$	Divide by the number left over.

ANSWER **45**

Type 2: finding the whole

EXAMPLE 3.2

150 is 25% of what number?

$\frac{150}{x} = \frac{25}{100}$	150 goes with *is*: the part. "What number" follows *of*: the whole is unknown.
$150 \times 100 = 15{,}000$	Cross-multiply.
$15{,}000 \div 25 = 600$	Divide by the number left over.

ANSWER **600**

Type 3: finding the percent

EXAMPLE 3.3

50 is what percent of 800?

$\frac{50}{800} = \frac{x}{100}$	50 is the part, 800 follows *of*, and the percent is unknown.
$50 \times 100 = 5000$	Cross-multiply.
$5000 \div 800 = 6.25$	Divide by the number left over.
6.25%	The answer sits in the percent position, so it already is a percent. Just add the sign.

ANSWER **6.25%**

CALCULATOR TIP **One line on the calculator**

Each example is a single chain: `25 × 180 ÷ 100`, `150 × 100 ÷ 25`, `50 × 100 ÷ 800`. Multiply the diagonal pair, divide by the third number. No parentheses are needed.

Table 3.1 The three types at a glance

The question	Proportion	Keystrokes
What is P% of W?	$\frac{x}{W} = \frac{P}{100}$	P × W ÷ 100
A is P% of what number?	$\frac{A}{x} = \frac{P}{100}$	A × 100 ÷ P
A is what percent of W?	$\frac{A}{W} = \frac{x}{100}$	A × 100 ÷ W

WATCH OUT **Check that the answer makes sense**

25% is one quarter. One quarter of 180 should be less than 180, and 45 is. If the percent is under 100, the part must be smaller than the whole. If your answer breaks that rule, two numbers are in the wrong seats.

PRACTICE 3.1 **The three types**

Set up the percent proportion for each and solve. Do not round.

1. What is 30% of 400?
2. 30 is what percent of 400?
3. 400 is 80% of what number?
4. What is 15% of 60?
5. 18 is 12% of what number?
6. 9 is what percent of 72?
7. What is 6.5% of 240?
8. 45 is what percent of 36?
9. 21 is 3.5% of what number?
10. What is 120% of 85?
11. 3 is what percent of 480?
12. 7.2 is 16% of what number?
13. What is 0.5% of 900?
14. 66 is what percent of 240?
15. 250 is 125% of what number?
16. What is 37.5% of 64?

3.3 The decimal shortcut

Since percent means "÷ 100," the proportion can be shortened by changing the percent to a decimal first. These three rules give exactly the same answers as the proportion. Use whichever method you trust most.

KEY RULE **Percent problems with decimals**

Part = whole × percent (as a decimal) 25% of 180 = 180 × 0.25 = 45

Whole = part ÷ percent (as a decimal) 150 ÷ 0.25 = 600

Percent = part ÷ whole, then move the point 2 places right 50 ÷ 800 = 0.0625 = 6.25%

3.4 Percent on the job

Word problems hide the words *is* and *of*. Find the three pieces by asking: What is the total or original amount (the whole)? What piece of it is being described (the part)? Which number is the percent?

EXAMPLE 3.4 Efficiency

An air conditioner draws 3,500 watts of electrical power and delivers 2,870 watts of useful work at the compressor. What is its efficiency as a percent?

whole = 3,500 W; part = 2,870 W	The power going in is the total. The useful output is the part of it that is not lost.
$\frac{2870}{3500} = \frac{x}{100}$	Efficiency asks: the output is what percent of the input?
2870 × 100 ÷ 3500 = 82	Cross-multiply and divide.

ANSWER **82% efficient**

EXAMPLE 3.5 Discount

A roll of copper tubing lists for $186.00. The supply house gives contractors a 15% discount. What is the contractor price?

$\frac{x}{186} = \frac{15}{100}$	The discount is 15% **of** the list price. The list price is the whole.
15 × 186 ÷ 100 = 27.90	The discount is $27.90.
186.00 − 27.90 = 158.10	The question asks for the price, not the discount. Subtract.

ANSWER **$158.10**

A shortcut for the last example: if 15% comes off, 85% is left. So the price is 85% of 186, or 186 × 0.85 = 158.10 in one step.

EXAMPLE 3.6 Waste allowance

A sheet-metal job needs 240 square feet of material. The shop adds 12% for waste. How much should be ordered?

$12 \times 240 \div 100 = 28.8$	The waste allowance is 12% of the amount needed.
$240 + 28.8 = 268.8$	Add the allowance to the original amount.

ANSWER **268.8 sq ft**

EXAMPLE 3.7 Working backward

After a 6% sales tax, a tool costs $84.80. What was the price before tax?

$100\% + 6\% = 106\%$	The total is the original price **plus** 6% of it, so \$84.80 is 106% of the price.
$\frac{84.80}{x} = \frac{106}{100}$	84.80 is the part (it *is* 106% *of* the price). The price is the whole.
$84.80 \times 100 \div 106 = 80$	Cross-multiply and divide.

ANSWER **$80.00**

WATCH OUT **Do not take the percent of the wrong number**

In the last example, 6% of \$84.80 is \$5.09, and 84.80 − 5.09 = \$79.71, which is wrong. The tax was 6% of the *original* price, not of the total. When the whole is the unknown, set up the proportion; do not guess with subtraction.

Percent change

A percent increase or decrease always compares the **amount of change** to the **original** value.

KEY RULE **Percent change**

$$\text{percent change} = \frac{\text{amount of change}}{\text{original amount}} \times 100$$

The original amount is the whole, no matter whether the value went up or down.

EXAMPLE 3.8

A supply-air temperature drops from 75°F to 57°F. What is the percent decrease?

$75 - 57 = 18$	Find the amount of change.
$\frac{18}{75} = \frac{x}{100}$	The change is the part. The **original** value is the whole.
$18 \times 100 \div 75 = 24$	Cross-multiply and divide.

ANSWER **24% decrease**

PRACTICE 3.2 **Applications**

Solve each problem. Round only where the problem says.

1. A 250-ft spool of wire has had 85 ft used. What percent has been used?
2. A motor rated at 1,200 W input is 88% efficient. What is its output power?
3. A technician earns $24.50 per hour and receives a 6% raise. What is the new hourly rate, to the nearest cent?
4. A furnace listed at $1,450 is sold at a 20% discount. What is the sale price?
5. A job needs 320 sq ft of drywall. Add 10% for waste. How much is ordered?
6. 18 of 150 welds failed inspection. What percent failed? What percent passed?
7. A down payment of $1,260 is 15% of the price of a used service van. What is the price?
8. A tank holds 60 gallons when it is 75% full. What is its full capacity?
9. Line voltage falls from 240 V to 228 V. What is the percent decrease?
10. The price of a fitting rises from $8.00 to $9.20. What is the percent increase?
11. A resistor marked 470 Ω has a tolerance of ±5%. What are its lowest and highest acceptable values?
12. With 8% sales tax a compressor costs $702. What was the price before tax?

CHAPTER SUMMARY

- Every percent problem fits one sentence: the **part** *is* a **percent** *of* the **whole**.
- The percent proportion is part over whole equals percent over 100.
- Cross-multiply the diagonal pair of numbers, then divide by the third number.
- Decimal shortcut: part = whole × percent; whole = part ÷ percent; percent = part ÷ whole.
- Percent change compares the amount of change to the **original** amount.

CHAPTER 3 REVIEW

1. What is 35% of 260?
2. 84 is 70% of what number?
3. 27 is what percent of 180?
4. What is 4.5% of 800?
5. 150 is what percent of 120?
6. 9 is 0.6% of what number?
7. What is the whole in the statement “12 is 40% of 30”?
8. Write the proportion for “What percent of 64 is 24?” Do not solve.
9. Use the decimal shortcut: find 18% of 350.
10. Use the decimal shortcut: 91 is 65% of what number?
11. A pump delivers 1,530 W of output for 1,800 W of input. What is its efficiency?
12. A $640 meter is on sale for 25% off. What is the sale price?
13. Static pressure rises from 0.50 to 0.62 inch of water. What is the percent increase?
14. A crew has installed 45 of 125 light fixtures. What percent remain to be installed?
15. A bill of $1,391 includes 7% sales tax. What was the amount before tax?
16. A student says 8% of 50 is 40. Explain the error and give the correct answer.

APPLIED TECHNICAL MATH

Part 2 · Measurement

Measuring is counting units. Part 2 covers the two systems of units used in American shops, how to move between them, and how to read the instruments that do the measuring.

CHAPTER 4

Standard Units of Measure

Carpet is sold by the square yard, concrete by the cubic yard, ductwork is laid out in inches and a tank is rated in gallons. The measurements on a job rarely arrive in the unit you need. This chapter is about changing the unit without changing the amount.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- State the standard (U.S. customary) units of length, weight and liquid capacity and how they relate.
- Convert a measurement from one unit to another by multiplying or dividing by a conversion factor.
- Decide whether to multiply or divide.
- Convert square units of area and cubic units of volume.
- Convert between cubic measure and gallons.

KEY TERMS

unit	A fixed amount used for measuring, such as a foot or a pound.
conversion factor	The number of smaller units that make one larger unit: 12 inches make 1 foot.
square unit	A unit of area: a square that is one unit long on each side.
cubic unit	A unit of volume: a cube that is one unit long on each edge.
capacity	How much liquid a container holds.

4.1 The standard units

The standard system, also called the U.S. customary or English system, grew out of everyday objects and trade. Its conversion factors are not regular, so they have to be memorised or looked up.

Table 4.1 Standard units of length, weight and capacity

Length	Weight	Liquid capacity
1 foot (ft) = 12 inches (in)	1 pound (lb) = 16 ounces (oz)	1 pint (pt) = 2 cups
1 yard (yd) = 3 ft = 36 in	1 ton (T) = 2000 lb	1 quart (qt) = 2 pt
1 mile (mi) = 5280 ft = 1760 yd		1 gallon (gal) = 4 qt

A measurement has two parts, a **number** and a **unit**. Converting changes both: the same board is 8 feet or 96 inches. The number grew because the unit shrank.

This chapter takes the standard system one **category** at a time. Each category has its own set of facts:

Table 4.2 The five conversion categories

Category	What it measures	Units	Section
Length	how long, how far	in, ft, yd, mi	4.3
Weight	how heavy	oz, lb, T	4.4
Liquid capacity	how much a container holds	cup, pt, qt, gal	4.5
Area	how much surface	sq in, sq ft, sq yd	4.6
Volume	how much space	cu in, cu ft, cu yd, and gallons	4.7

4.2 Multiply or divide?

KEY RULE **Converting units**

Going to a **smaller** unit, you will need **more** of them: **multiply** by the conversion factor.

Going to a **larger** unit, you will need **fewer** of them: **divide** by the conversion factor.

EXAMPLE 4.1

Convert 7.5 ft to inches.

1 ft = 12 in	Find the fact that connects the two units.
feet → inches: larger → smaller	Inches are smaller than feet, so there will be more of them. Multiply.
7.5 × 12 = 90	Multiply by the conversion factor.

ANSWER **90 in**

EXAMPLE 4.2

Convert 150 oz to pounds. Round to the nearest tenth.

1 lb = 16 oz	The connecting fact.
ounces → pounds: smaller → larger	Pounds are larger, so there will be fewer. Divide.
150 ÷ 16 = 9.375	Divide by the conversion factor.
9.4	Round to the nearest tenth: the test digit 7 rounds the 3 up.

ANSWER **9.4 lb**

WATCH OUT **The sanity check**

Before writing the answer, ask whether the number should have gone up or down. 150 ounces cannot be 2400 pounds. If the answer moved the wrong way, you multiplied when you should have divided.

Conversions that take two steps

When no single fact joins the two units, go through a unit in between, one step at a time.

EXAMPLE 4.3

How many cups are in 3 gallons?

3 × 4 = 12 qt	Gallons to quarts: smaller unit, multiply by 4.
12 × 2 = 24 pt	Quarts to pints: multiply by 2.
24 × 2 = 48 cups	Pints to cups: multiply by 2.

ANSWER **48 cups**

Unit fractions: a method that cannot go the wrong way

A conversion fact can be written as a fraction equal to 1, such as $\frac{12\text{ in}}{1\text{ ft}}$ or $\frac{1\text{ ft}}{12\text{ in}}$. Multiplying by 1 does not change an amount. Choose the fraction that has the unit you want to **get rid of on the bottom**, so that it cancels.

> 7.5 ft × $\frac{12\text{ in}}{1\text{ ft}}$ = 90 in (ft cancels ft, inches remain)
>
> 150 oz × $\frac{1\text{ lb}}{16\text{ oz}}$ = 9.375 lb (oz cancels oz, pounds remain)

4.3 Category 1: length

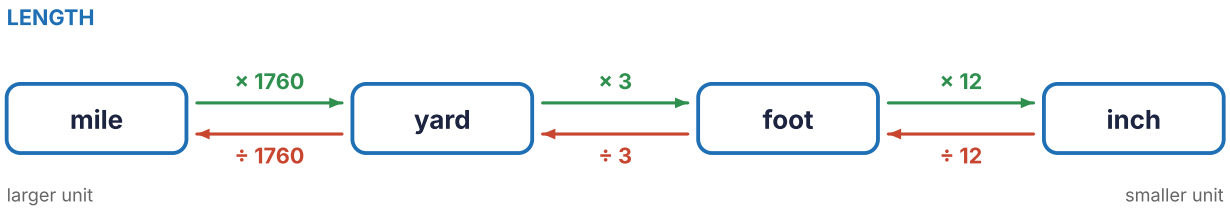


Figure 4.1 The length chain. Move right (to a smaller unit) and multiply; move left (to a larger unit) and divide.

Fact	To go to the smaller unit	To go to the larger unit
1 ft = 12 in	ft → in: × 12	in → ft: ÷ 12
1 yd = 3 ft	yd → ft: × 3	ft → yd: ÷ 3
1 yd = 36 in	yd → in: × 36	in → yd: ÷ 36
1 mi = 5280 ft	mi → ft: × 5280	ft → mi: ÷ 5280
1 mi = 1760 yd	mi → yd: × 1760	yd → mi: ÷ 1760

EXAMPLE 4.4

A run of conduit is 46 ft long. How many yards is that, to the nearest tenth? How many inches?

46 ÷ 3 = 15.333…	Feet to yards: a yard is larger, so divide by 3.
15.3 yd	Round to the nearest tenth.
46 × 12 = 552 in	Feet to inches: an inch is smaller, so multiply by 12.

ANSWER **15.3 yd; 552 in**

EXAMPLE 4.5

A pipeline is 3960 ft long. What part of a mile is that?

1 mi = 5280 ft	The fact.
3960 ÷ 5280 = 0.75	Feet to miles: divide.

ANSWER **0.75 mi**

4.4 Category 2: weight



Figure 4.2 The weight chain.

Fact	To go to the smaller unit	To go to the larger unit
1 lb = 16 oz	lb → oz: × 16	oz → lb: ÷ 16
1 T = 2000 lb	T → lb: × 2000	lb → T: ÷ 2000

EXAMPLE 4.6

A rooftop unit weighs 1.75 tons. How many pounds is that?

1 T = 2000 lb	The fact.
1.75 × 2000 = 3500	Tons to pounds: smaller unit, multiply.

ANSWER **3500 lb**

EXAMPLE 4.7 A measurement in two units

A system is charged with 6 lb 4 oz of refrigerant. Write the charge in ounces, and in pounds as a decimal.

6 × 16 = 96 oz	Change the pounds to ounces.
96 + 4 = 100 oz	Add the extra ounces.
4 ÷ 16 = 0.25 lb	Or change the 4 ounces to pounds.
6 + 0.25 = 6.25 lb	Add to the whole pounds.

ANSWER **100 oz, or 6.25 lb**

4.5 Category 3: liquid capacity

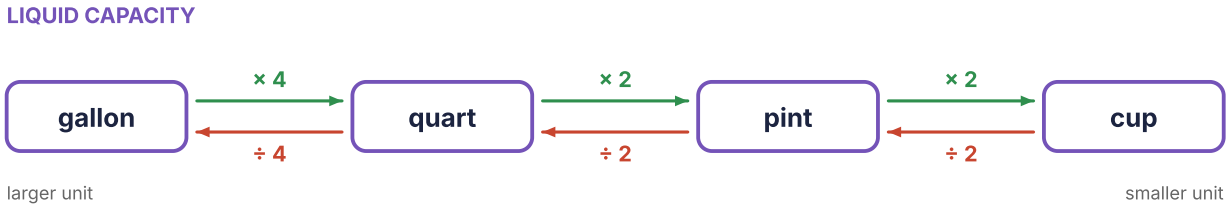


Figure 4.3 The capacity chain.

Fact	To go to the smaller unit	To go to the larger unit
1 pt = 2 cups	pt → cups: × 2	cups → pt: ÷ 2
1 qt = 2 pt	qt → pt: × 2	pt → qt: ÷ 2
1 gal = 4 qt	gal → qt: × 4	qt → gal: ÷ 4
1 gal = 8 pt = 16 cups	gal → pt: × 8	pt → gal: ÷ 8

EXAMPLE 4.8

A compressor takes 54 fluid quarts of oil over a year. How many gallons is that?

1 gal = 4 qt	The fact.
54 ÷ 4 = 13.5	Quarts to gallons: larger unit, divide.

ANSWER **13.5 gal**

EXAMPLE 4.9

How many pints are in 2.5 gallons?

2.5 × 4 = 10 qt	Gallons to quarts.
10 × 2 = 20 pt	Quarts to pints.

ANSWER **20 pt**

PRACTICE 4.1 **Length, weight and capacity**

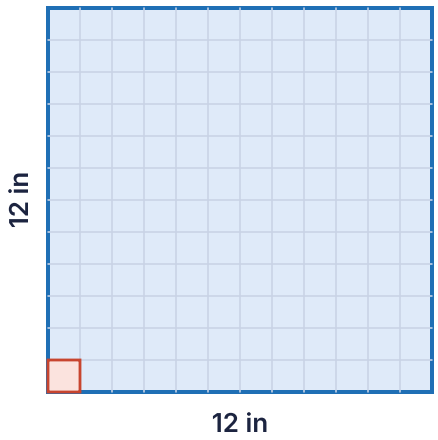
Convert. Round to the nearest tenth if needed. Problems 1 to 6 are length, 7 to 10 weight, 11 to 14 capacity, 15 and 16 length.

1. 9 ft to in
2. 102 in to ft
3. 4.5 yd to ft
4. 50 ft to yd
5. 2.5 mi to ft
6. 9,000 ft to mi

7. 5 lb to oz
8. 72 oz to lb
9. 3.2 T to lb
10. 5,500 lb to T
11. 6 gal to qt
12. 18 qt to gal
13. 7 pt to qt
14. 2 gal to pt
15. 180 in to yd
16. 0.75 mi to yd

4.6 Category 4: area

Area is measured in **square units**. A square foot is a square that measures 1 foot on each side. Since each side is 12 inches, the square holds 12 rows of 12 square inches: 144 of them, not 12.



1 square foot
= 12 in × 12 in
= 144 square inches

Length factor 12, so the area factor is 12 × 12.

Figure 4.4 One square foot holds 144 square inches. The small red square is one square inch.

KEY RULE **Area conversion factors**

Square the length factor.

1 sq ft = 12 × 12 = **144** sq in

1 sq yd = 3 × 3 = **9** sq ft

1 sq yd = 36 × 36 = **1296** sq in

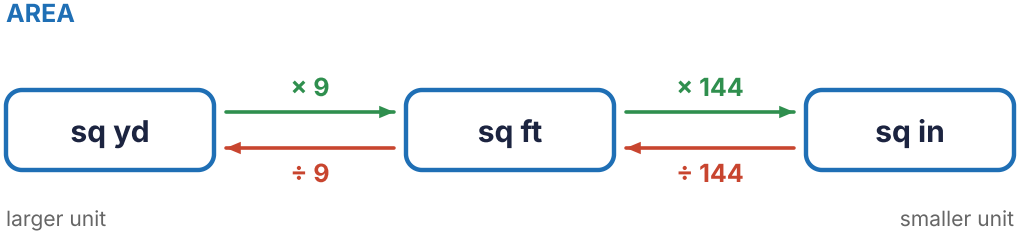


Figure 4.5 The area chain. From square yards straight to square inches the factor is 9 × 144 = 1296.

EXAMPLE 4.10

A duct opening has an area of 1000 sq in. How many square feet is that, to the nearest tenth?

1 sq ft = 144 sq in	The area fact, not the length fact.
sq in → sq ft: smaller → larger	Divide.
1000 ÷ 144 = 6.944…	Divide by 144.
6.9	Round to the nearest tenth.

ANSWER **6.9 sq ft**

EXAMPLE 4.11

A room needs 15.5 sq yd of carpet. How many square feet is that?

1 sq yd = 9 sq ft	The area fact.
15.5 × 9 = 139.5	Larger unit to smaller: multiply.

ANSWER **139.5 sq ft**

WATCH OUT **The most common conversion mistake**

Dividing square inches by 12 to get square feet. Twelve converts *lengths*. Areas use 144. Watch for the words "square" or "sq" or the exponent in in^2 and ft^2.

4.7 Category 5: volume

Volume is measured in **cubic units**. A cubic foot is a cube 1 foot long on every edge. It holds 12 layers, each with 144 cubic inches, so the length factor is used **three** times.

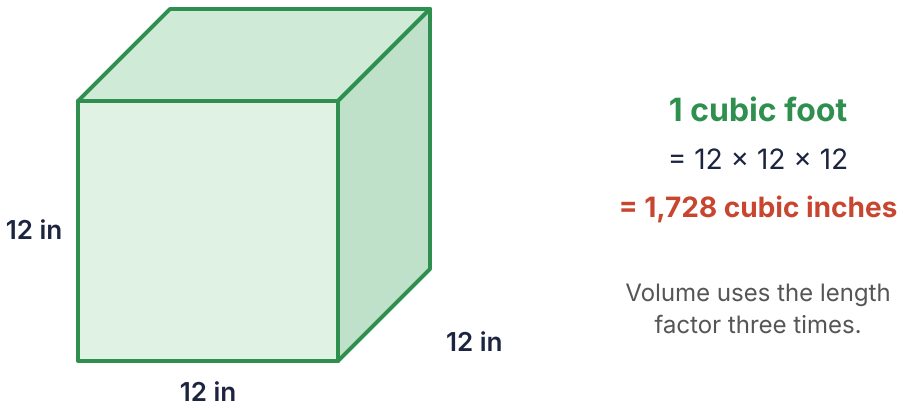


Figure 4.6 One cubic foot holds 12 × 12 × 12 = 1,728 cubic inches.

KEY RULE **Volume conversion factors**

Cube the length factor.

1 cu ft = 12 × 12 × 12 = **1728** cu in

1 cu yd = 3 × 3 × 3 = **27** cu ft

1 cu yd = 36 × 36 × 36 = **46,656** cu in

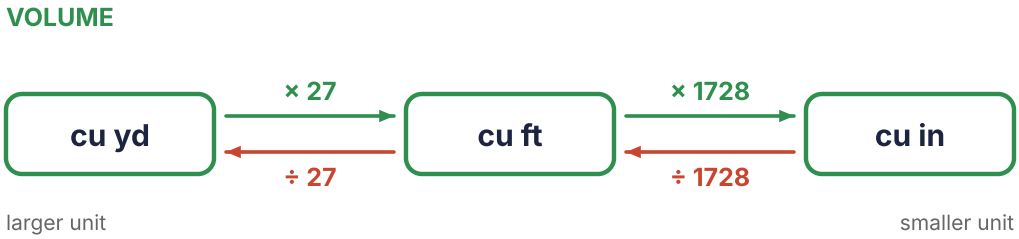


Figure 4.7 The volume chain. From cubic yards straight to cubic inches the factor is 27 × 1728 = 46,656.

EXAMPLE 4.12

An air handler cabinet has a volume of 5.5 cu ft. How many cubic inches is that?

1 cu ft = 1728 cu in	The volume fact.
5.5 × 1728 = 9504	Larger unit to smaller: multiply.

ANSWER **9504 cu in**

EXAMPLE 4.13

A slab needs 2500 cu ft of concrete. Concrete is ordered by the cubic yard. How many cubic yards, to the nearest tenth?

1 cu yd = 27 cu ft	The volume fact.
2500 ÷ 27 = 92.592…	Smaller unit to larger: divide.
92.6	Round to the nearest tenth.

ANSWER **92.6 cu yd**

Cubic measure and gallons

Tanks, pans and sumps are built to dimensions in inches or feet but are filled in gallons. Two facts connect them.

KEY RULE **Volume and liquid capacity**

1 cu ft = **7.48** gallons

1 gallon = **231** cu in



Figure 4.8 A gallon sits between the cubic foot and the cubic inch.

EXAMPLE 4.14

A condensate tank has a volume of 5 cu ft. How many gallons does it hold?

1 cu ft = 7.48 gal	A gallon is smaller than a cubic foot, so there will be more gallons.
5 × 7.48 = 37.4	Multiply.

ANSWER **37.4 gal**

EXAMPLE 4.15

A drain pan holds 1500 cu in. How many gallons is that, to the nearest tenth?

1 gal = 231 cu in	A gallon is larger than a cubic inch.
1500 ÷ 231 = 6.493…	Divide.
6.5	Round.

ANSWER **6.5 gal**

Table 4.3 Area and volume factors. Multiply going to the smaller unit; divide going to the larger.

To convert	Factor	To convert	Factor
sq ft ↔ sq in	144	cu ft ↔ cu in	1728
sq yd ↔ sq ft	9	cu yd ↔ cu ft	27
sq yd ↔ sq in	1296	cu yd ↔ cu in	46,656
cu ft ↔ gal	7.48	gal ↔ cu in	231

ON THE JOB **Ordering concrete**

A patio 20 ft long, 12 ft wide and 4 in thick has a volume of $20 \times 12 \times \frac{4}{12} = 80$ cu ft. Divide by 27: 2.96 cu yd. The thickness had to be changed to feet first. All three dimensions must be in the same unit before multiplying.

PRACTICE 4.2 **Area and volume**

Convert. Round to the nearest tenth if needed.

1. 850 sq in to sq ft
2. 12.5 sq yd to sq ft
3. 18 sq ft to sq in
4. 2,000 sq in to sq yd
5. 400 sq ft to sq yd
6. 1,800 cu ft to cu yd
7. 4,000 cu in to cu ft
8. 16.5 cu yd to cu ft
9. 3.5 cu ft to cu in
10. 8 cu ft to gal
11. 2,000 cu in to gal
12. 40 gal to cu ft
13. 3 gal to cu in
14. 2 cu yd to cu in
15. 120 gal to cu ft
16. 0.5 sq yd to sq in

CHAPTER SUMMARY

- Larger unit to smaller unit: multiply. Smaller unit to larger unit: divide.
- Length facts: 12 in per ft, 3 ft per yd, 36 in per yd, 5280 ft per mile.
- Area factors are length factors squared: 144, 9, 1296.
- Volume factors are length factors cubed: 1728, 27, 46,656.
- 1 cu ft = 7.48 gal and 1 gal = 231 cu in.
- Check every answer: should the number have become larger or smaller?

CHAPTER 4 REVIEW

1. 14 ft to in
2. 200 in to ft
3. 7 lb to oz
4. 7,000 lb to T
5. 10 qt to gal
6. 600 sq in to sq ft
7. 25 sq yd to sq ft
8. 500 cu ft to cu yd
9. 2.5 cu ft to cu in
10. 12 cu ft to gal
11. 1,000 cu in to gal
12. 75 gal to cu ft

13. Why is the factor from square feet to square inches 144 and not 12?
14. A floor is 270 sq ft. Tile is sold by the square yard. How many square yards are needed?
15. A trench needs 135 cu ft of gravel. How many cubic yards is that?
16. A rectangular tank has a volume of 30 cu ft. How many gallons does it hold?
17. How many inches are in 2 miles?
18. A 55-gallon drum has what volume in cubic inches?

CHAPTER 5

The Metric System

Refrigerant charge in grams, pipe in millimetres, pressure in kilopascals: metric units appear on nearly every piece of modern equipment. The good news is that the metric system was designed to be easy. Every conversion inside it is a move of the decimal point.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Name the metric base units and the six common prefixes in order.
- Convert within the metric system by moving the decimal point.
- Convert between metric and standard units with the class conversion factors.
- Judge which of two units is larger.
- Carry out conversions that need more than one step.

KEY TERMS

base unit	The starting unit for a kind of measurement: metre (length), gram (weight), litre (capacity).
prefix	A word part put in front of the base unit that multiplies it by a power of ten.
metric stairs	The seven prefixes written in order from kilo down to milli; also called the metric ladder.

5.1 Base units and prefixes

The metric system has one base unit for each kind of measurement, and the same set of prefixes works with all of them.

Measures	Base unit	Symbol	About the size of
length	metre (meter)	m	a little more than a yard
weight (mass)	gram	g	a paper clip
liquid capacity	litre (liter)	L	a little more than a quart

Table 5.1 The metric prefixes, largest to smallest

Prefix	Symbol	Means	Length	Weight	Capacity
kilo	k	1000	km	kg	kL
hecto	h	100	hm	hg	hL
deka	da	10	dam	dag	daL
(base unit)		1	m	g	L
deci	d	0.1 ($\frac{1}{10}$)	dm	dg	dL
centi	c	0.01 ($\frac{1}{100}$)	cm	cg	cL
milli	m	0.001 ($\frac{1}{1000}$)	mm	mg	mL

CALCULATOR TIP **Remembering the order**

A sentence whose words start with **k h da (unit) d c m** keeps the prefixes in order. A common one: **K**ing **H**enry **D**ied **U**nusually **D**rinking **C**hocolate **M**ilk.

5.2 Converting on the metric stairs

Each step on the stairs is a factor of 10, and multiplying or dividing by 10 moves the decimal point one place. So a metric conversion is done by counting steps.

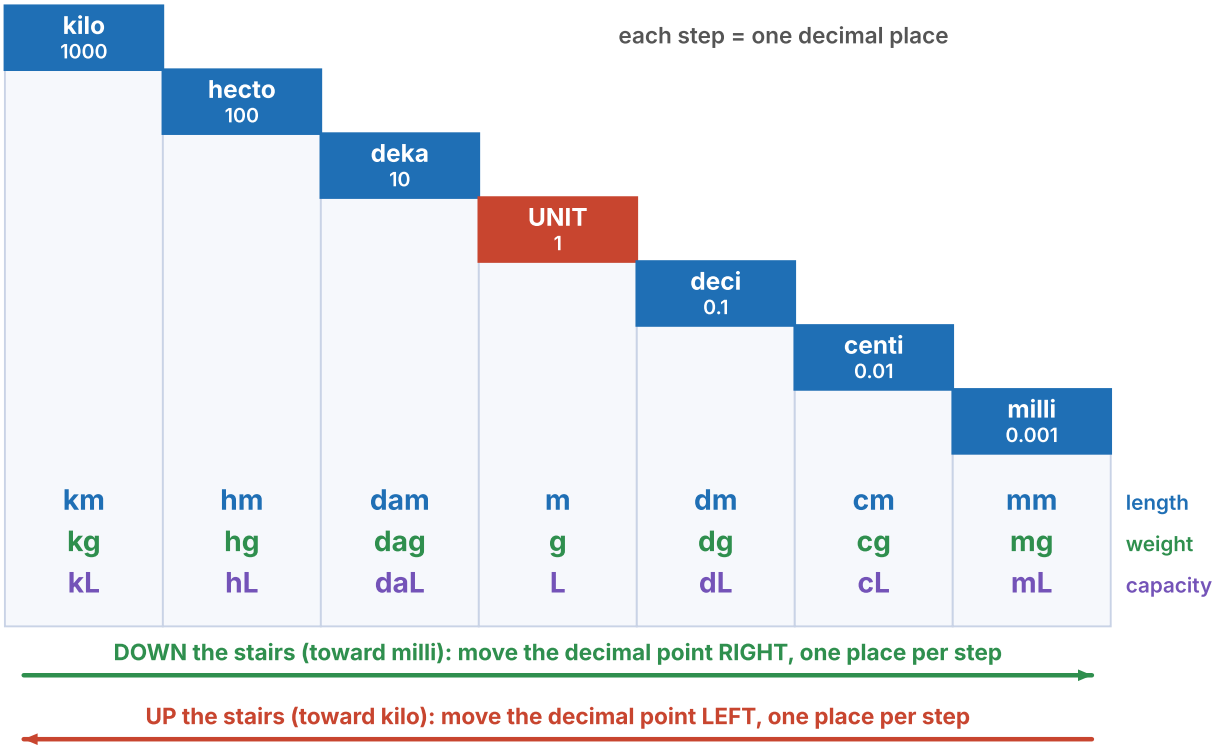


Figure 5.1 The metric stairs. Every step holds a length unit, a weight unit and a capacity unit, and the same stairs serve all three.

KEY RULE **Converting within the metric system**

1. Find the unit you have and the unit you want on the stairs.

2. Count the steps between them.

3. Move the decimal point that many places, **in the same direction you moved on the stairs**: going **down** toward milli (right on the chart), move the point **right**; going **up** toward kilo, move it **left**.

Fill empty places with zeros.

EXAMPLE 5.1

Convert 250 mm to centimetres.

mm → cm	Centi is one step **up** the stairs from milli (one step left on the chart).
250. → 25.0	Move the decimal point one place left.

ANSWER **25 cm**

EXAMPLE 5.2

Convert 0.25 L to millilitres.

L → dL → cL → mL	From the base unit to milli is three steps **down**.
0.25 → 0 250.	Move the point three places right. A zero must be added to fill the third place.

ANSWER **250 mL**

EXAMPLE 5.3

Convert 250 dg to dekagrams.

dg → g → dag	Deci to deka is two steps **up**, passing the base unit.
250. → 2.50	Move the point two places left.

ANSWER **2.5 dag**

WATCH OUT **Count steps, not units**

From mm to m is **three** steps (mm → cm → dm → m), even though only two units are named in the problem. Point at each stair as you count. And remember the unit step: deka → base → deci is two moves.

WATCH OUT **m, mm and M**

Lowercase **m** alone is metre. As a prefix, lowercase **m** is milli (mm, mg, mL). Do not confuse millimetres with metres, or mg with g: the difference is a factor of 1000.

5.3 The stairs, one category at a time

The stairs never change. Only the base unit changes with what is being measured.

Category 1: metric length (the metre)

km	hm	dam	m	dm	cm	mm
kilometre	hectometre	dekametre	**metre**	decimetre	centimetre	millimetre
1000 m	100 m	10 m	1 m	0.1 m	0.01 m	0.001 m

The units used most are the **kilometre** (road distances), the **metre** (rooms and pipe runs), the **centimetre** and the **millimetre** (parts and drawings). Worth knowing by heart: 1 m = 100 cm = 1000 mm, 1 cm = 10 mm, 1 km = 1000 m.

EXAMPLE 5.4

A pipe is 3.2 m long. Give its length in centimetres and in millimetres.

m → dm → cm: 2 steps down	Move the point 2 places right.
3.2 → 320 cm	
m → dm → cm → mm: 3 steps down	Move the point 3 places right.
3.2 → 3200 mm	

ANSWER **320 cm; 3200 mm**

EXAMPLE 5.5

Convert 4750 m to kilometres.

m → dam → hm → km: 3 steps up	Move the point 3 places left.
4750. → 4.750	

ANSWER **4.75 km**

Category 2: metric weight (the gram)

kg	hg	dag	g	dg	cg	mg
kilogram	hectogram	dekagram	**gram**	decigram	centigram	milligram
1000 g	100 g	10 g	1 g	0.1 g	0.01 g	0.001 g

The **kilogram**, the **gram** and the **milligram** do nearly all the work: 1 kg = 1000 g and 1 g = 1000 mg.

EXAMPLE 5.6

A refrigerant charge is 1.36 kg. How many grams is that?

kg → hg → dag → g: 3 steps down	Move the point 3 places right.
1.36 → 1360	Fill the empty place with a zero.

ANSWER **1360 g**

EXAMPLE 5.7

Convert 85 cg to grams.

cg → dg → g: 2 steps up	Move the point 2 places left.
85. → 0.85	

ANSWER **0.85 g**

Category 3: metric capacity (the litre)

kL	hL	daL	L	dL	cL	mL
kilolitre	hectolitre	dekalitre	**litre**	decilitre	centilitre	millilitre
1000 L	100 L	10 L	1 L	0.1 L	0.01 L	0.001 L

The **litre** and the **millilitre** are the everyday units: 1 L = 1000 mL.

EXAMPLE 5.8

A sample bottle holds 475 mL. How many litres is that?

mL → cL → dL → L: 3 steps up	Move the point 3 places left.
475. → 0.475	

ANSWER **0.475 L**

EXAMPLE 5.9

Convert 2.6 daL to decilitres.

daL → L → dL: 2 steps down	Deka to deci passes the base unit: two steps. Move the point 2 places right.
2.6 → 260	

ANSWER **260 dL**

The larger prefixes and the smaller ones

Science and electronics reach beyond the seven stairs. These prefixes step by a factor of 1000, three decimal places at a time.

Prefix	Symbol	Multiplies the unit by	Example
tera	T	1,000,000,000,000	terabyte
giga	G	1,000,000,000	gigawatt
mega	M	1,000,000	megohm (MΩ)
kilo	k	1,000	kilowatt (kW)
milli	m	0.001	milliamp (mA)
micro	µ	0.000001	microfarad (µF)
nano	n	0.000000001	nanosecond
pico	p	0.000000000001	picofarad (pF)

PRACTICE 5.1 **Metric to metric**

Convert by moving the decimal point.

1. 4.5 m to cm
2. 380 mm to cm
3. 2.4 km to m
4. 750 g to kg
5. 0.6 L to mL
6. 1,250 mL to L
7. 35 cm to mm
8. 8,200 m to km
9. 460 dg to dag
10. 3.2 hL to L
11. 0.085 kg to g
12. 95 cm to m
13. 7 dam to dm
14. 12,500 mg to g
15. 0.4 m to mm
16. 64 cL to L

5.4 Between metric and standard

Moving between the two systems needs conversion factors again. The course uses the seven below. Each line is a fact, and the multiply-or-divide rule from Chapter 4 applies: toward the **smaller** unit multiply, toward the **larger** unit divide.

Table 5.2 Standard and metric conversion facts used in this course

Length	Weight	Capacity
1 in = 2.54 cm	1 oz = 28.35 g	1 L = 1.06 qt
1 m = 3.28 ft	1 kg = 2.205 lb	1 gal = 3.785 L
1 mi = 1.61 km		

KEY RULE **Which unit is larger?**

In each fact, the unit with the **1** in front of it is the larger unit, because it takes more than one of the other to match it.

An inch is larger than a centimetre. A metre is larger than a foot (and a yard). A mile is larger than a kilometre. An ounce is larger than a gram. A kilogram is larger than a pound. A litre is larger than a quart. A gallon is larger than a litre.

EXAMPLE 5.10

Convert 50 cm to inches. Round to the nearest tenth.

1 in = 2.54 cm	The connecting fact. The inch is the larger unit.
cm → in: smaller → larger	Divide.
50 ÷ 2.54 = 19.685...	Divide by the factor.
19.7	Round.

ANSWER **19.7 in**

EXAMPLE 5.11

Convert 185 lb to kilograms. Round to the nearest tenth.

1 kg = 2.205 lb	The kilogram is the larger unit.
lb → kg: smaller → larger	Divide.
185 ÷ 2.205 = 83.900...	Divide.
83.9	Round.

ANSWER **83.9 kg**

EXAMPLE 5.12

Convert 50 L to quarts.

1 L = 1.06 qt	The litre is the larger unit.
L → qt: larger → smaller	Multiply.
50 × 1.06 = 53	Multiply.

ANSWER **53 qt**

The three examples above come from the three categories. Each category is now set out in full, in both directions.

Category 1: length (inch and centimetre, metre and foot, mile and kilometre)

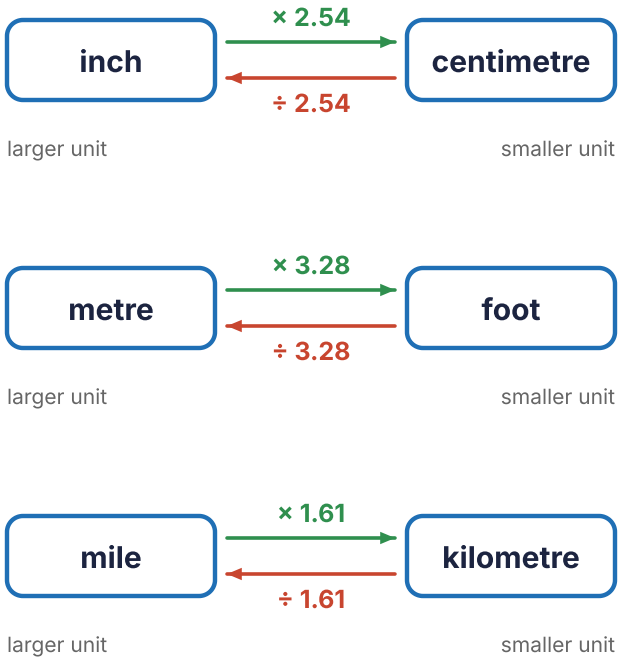


Figure 5.2 Length between the systems.

Fact	Multiply	Divide
1 in = 2.54 cm	in → cm: × 2.54	cm → in: ÷ 2.54
1 m = 3.28 ft	m → ft: × 3.28	ft → m: ÷ 3.28
1 mi = 1.61 km	mi → km: × 1.61	km → mi: ÷ 1.61

EXAMPLE 5.13

A room is 4.5 m long. How many feet is that, to the nearest tenth?

1 m = 3.28 ft	The metre is the larger unit.
4.5 × 3.28 = 14.76	Metres to feet: multiply.
14.8	Round.

ANSWER **14.8 ft**

EXAMPLE 5.14

A job site is 12 km away. How many miles is that, to the nearest tenth?

1 mi = 1.61 km	The mile is the larger unit.
12 ÷ 1.61 = 7.453…	Kilometres to miles: divide.
7.5	Round.

ANSWER **7.5 mi**

Category 2: weight (ounce and gram, kilogram and pound)

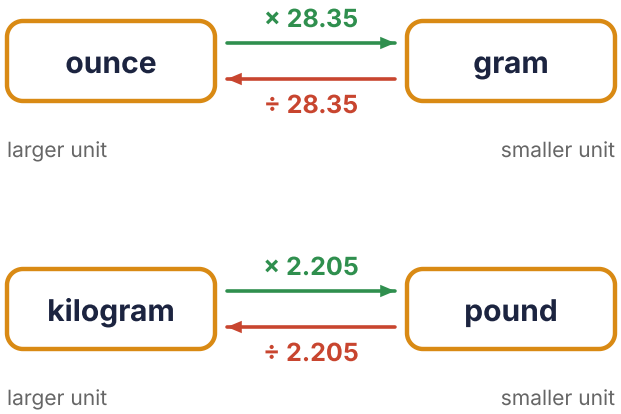


Figure 5.3 Weight between the systems.

Fact	Multiply	Divide
1 oz = 28.35 g	oz → g: × 28.35	g → oz: ÷ 28.35
1 kg = 2.205 lb	kg → lb: × 2.205	lb → kg: ÷ 2.205

EXAMPLE 5.15

A nameplate gives a charge of 14 oz. How many grams is that?

1 oz = 28.35 g	The ounce is the larger unit.
14 × 28.35 = 396.9	Ounces to grams: multiply.

ANSWER **396.9 g**

EXAMPLE 5.16

A motor has a mass of 32 kg. How many pounds is that, to the nearest tenth?

1 kg = 2.205 lb	The kilogram is the larger unit.
32 × 2.205 = 70.56	Kilograms to pounds: multiply.
70.6	Round.

ANSWER **70.6 lb**

Category 3: capacity (litre and quart, gallon and litre)

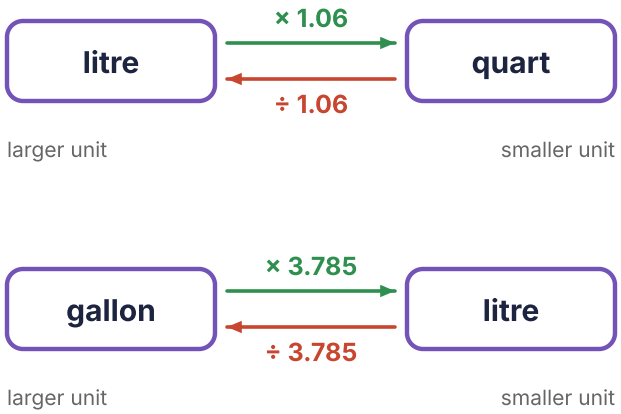


Figure 5.4 Capacity between the systems.

Fact	Multiply	Divide
1 L = 1.06 qt	L → qt: × 1.06	qt → L: ÷ 1.06
1 gal = 3.785 L	gal → L: × 3.785	L → gal: ÷ 3.785

EXAMPLE 5.17

A sump holds 12 gal. How many litres is that, to the nearest tenth?

1 gal = 3.785 L	The gallon is the larger unit.
12 × 3.785 = 45.42	Gallons to litres: multiply.
45.4	Round.

ANSWER **45.4 L**

EXAMPLE 5.18

An oil charge is 8 qt. How many litres is that, to the nearest tenth?

1 L = 1.06 qt	The litre is the larger unit.
8 ÷ 1.06 = 7.547…	Quarts to litres: divide.
7.5	Round.

ANSWER **7.5 L**

Conversions with more than one step

If the unit you have is not one of the units in the table, first move along the metric stairs or use a standard fact to reach a unit that is. Write down each step with its unit.

EXAMPLE 5.19

Convert 5 km to inches. Round to the nearest whole number.

5 km = 500,000 cm	The table joins inches to **centimetres**. From kilo to centi is five steps down: move the point five places right.
500,000 ÷ 2.54 = 196,850.39…	Centimetres to inches: smaller to larger, divide by 2.54.
196,850	Round to the nearest whole number.

ANSWER **196,850 in**

EXAMPLE 5.20

A recovery cylinder holds 30 lb of refrigerant. How many grams is that? Round to the nearest whole number.

30 ÷ 2.205 = 13.6054… kg	Pounds to kilograms: divide. Keep the digits in the calculator.
13.6054… kg = 13,605.4… g	Kilograms to grams: three steps down the stairs.
13,605	Round at the end.

ANSWER **13,605 g**

ON THE JOB **Why technicians need both systems**

Refrigerant charge on a nameplate may be given in ounces and in grams. Torque specs appear in foot-pounds and newton-metres. Imported equipment is dimensioned in millimetres, but the tape on your belt reads inches. A quick mental check helps: an inch is about 25 mm, a kilogram is a little over 2 lb, and a litre is about a quart.

PRACTICE 5.2 **Metric and standard**

Convert. Round to the nearest tenth if needed. For 13 to 16, circle (name) the larger unit.

1. 12 in to cm
2. 80 cm to in
3. 25 m to ft
4. 100 ft to m
5. 26.2 mi to km
6. 10 km to mi
7. 8 oz to g
8. 500 g to oz
9. 70 kg to lb
10. 220 lb to kg
11. 20 L to qt
12. 15 gal to L
13. centimetre or inch
14. litre or quart
15. gram or ounce
16. kilometre or mile

PRACTICE 5.3 **Conversions by category**

Convert. Round to the nearest tenth if needed. 1 to 6 length, 7 to 12 weight, 13 to 18 capacity.

1. 2.75 m to cm
2. 640 mm to m
3. 3.4 km to m
4. 18 in to cm
5. 60 ft to m
6. 55 mi to km
7. 2.2 kg to g
8. 350 mg to g
9. 48 g to cg
10. 24 oz to g
11. 1,000 g to oz
12. 45 kg to lb
13. 1.5 L to mL
14. 2,400 mL to L
15. 36 dL to L
16. 30 qt to L
17. 60 L to gal
18. 7.5 gal to L

CHAPTER SUMMARY

- Base units: metre, gram, litre. Prefixes in order: kilo, hecto, deka, (unit), deci, centi, milli.
- Inside the metric system, count the steps on the stairs and move the decimal point that many places, the same direction.
- Between systems use the facts 2.54, 3.28, 1.61, 28.35, 2.205, 1.06 and 3.785.
- The unit written with the 1 is the larger unit; multiply toward smaller units, divide toward larger ones.
- For multi-step conversions, write each step with its unit and round only at the end.

CHAPTER 5 REVIEW

1. 6.8 m to mm
2. 425 cm to m
3. 0.75 kg to g
4. 900 mL to L
5. 52 dg to dag
6. 1.2 km to cm
7. 36 in to cm
8. 150 cm to in
9. 5 kg to lb
10. 160 lb to kg
11. 40 L to qt
12. 5 gal to L
13. 26 qt to L
14. 400 km to mi
15. Which is larger, a metre or a yard?
16. Which is larger, a gallon or a cubic foot?
17. Convert 2 m to inches, to the nearest tenth. (Hint: go through centimetres.)
18. Convert 3 lb to grams, to the nearest whole number.
19. A pipe is 19 mm in diameter. What is that in inches, to the nearest hundredth?
20. How many steps, and in which direction, does the decimal point move from hectolitres to centilitres?

CHAPTER 6

Rulers, Gauges and Tolerance

A calculation is only as good as the measurement that goes into it. This chapter is about reading instruments correctly, and about deciding whether a measurement is close enough.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Read a metric rule in millimetres and centimetres.
- Read an inch rule to the nearest sixteenth and give the reading in lowest terms.
- Estimate the reading of a dial gauge, including a gauge with two scales.
- Find the upper and lower limits of a toleranced dimension and decide whether a part is acceptable.
- Find a missing dimension on a drawing.

KEY TERMS

graduation	One of the marks on a scale.
scale value	What the space between two neighbouring marks is worth.
tolerance	How far a dimension is allowed to vary from the size called for.
limits	The largest and smallest acceptable sizes.
nominal size	The size called for on the drawing, before the tolerance is applied.

6.1 The metric rule

On a metric rule the numbered marks are **centimetres**. Each centimetre is divided into 10 equal parts, and each small part is one **millimetre**. A slightly longer mark shows the halfway point, 5 mm.

KEY RULE **Reading a metric rule in millimetres**

Multiply the last numbered centimetre mark before the point by 10, then add the small marks past it.

A point 6 small marks past the 4 is 40 + 6 = 46 mm, which is also 4.6 cm.

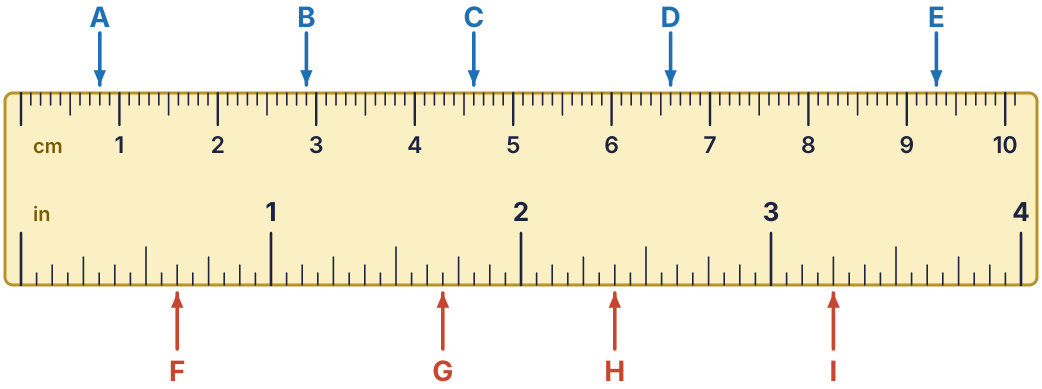


Figure 6.1 A rule with a centimetre scale on the top edge and an inch scale in sixteenths on the bottom edge.

EXAMPLE 6.1

Read points A to E on the metric scale of Figure 6.1, in millimetres.

A: 8 marks from zero	Before the 1 cm mark, so just count: 8 mm.
B: 2 cm + 9 marks = 29	One mark short of 3 cm.
C: 4 cm + 6 marks = 46	One mark past the long 5 mm mark.
D: 6 cm + 6 marks = 66	
E: 9 cm + 3 marks = 93	

ANSWER **A 8 mm, B 29 mm, C 46 mm, D 66 mm, E 93 mm**

6.2 The inch rule

An inch rule is divided by halving, again and again. Half of an inch is $\frac{1}{2}$, half of that is $\frac{1}{4}$, then $\frac{1}{8}$, then $\frac{1}{16}$. The marks are made in different lengths so the eye can pick them out: **the longer the mark, the larger the fraction.**

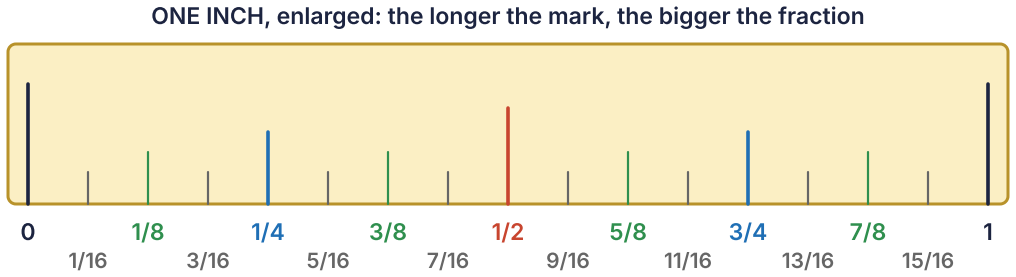


Figure 6.2 One inch divided into sixteenths. Every mark has a name in lowest terms.

KEY RULE **Reading an inch rule**

1. Write down the last whole inch before the point.

2. Count the sixteenth spaces from that inch mark to the point.

3. Write the count over 16 and reduce: keep halving top and bottom until the top is odd.

EXAMPLE 6.2

Read points F to I on the inch scale of Figure 6.1.

F: 0 in + 10 sixteenths = $\frac{10}{16} = \frac{5}{8}$	Ten spaces from zero. Reduce by 2.
G: 1 in + 11 sixteenths = $1\ \frac{11}{16}$	Eleven is odd, so it is already in lowest terms.
H: 2 in + 6 sixteenths = $2\ \frac{6}{16} = 2\ \frac{3}{8}$	Reduce by 2.
I: 3 in + 4 sixteenths = $3\ \frac{4}{16} = 3\ \frac{1}{4}$	Reduce by 4. This is the quarter-inch mark, the third-longest line.

ANSWER **F $\frac{5}{8}$ in, G $1\ \frac{11}{16}$ in, H $2\ \frac{3}{8}$ in, I $3\ \frac{1}{4}$ in**

CALCULATOR TIP **Count from the nearest long mark**

Counting eleven tiny spaces invites mistakes. Instead, find the nearest half-inch mark (that is $\frac{8}{16}$) and count from there: three marks past the half is $\frac{11}{16}$. One mark before the next inch is $\frac{15}{16}$.

WATCH OUT **Always reduce**

An answer of $2\ \frac{6}{16}$ is the right place on the ruler but is not finished. Readings are given in lowest terms: $2\ \frac{3}{8}$.

Rules with other divisions

Some rules are divided into 8ths, 32nds or 64ths, and engineers' and machinists' rules are divided into 10ths and 100ths. The method never changes: **count how many spaces make one inch**, and that count is the denominator.

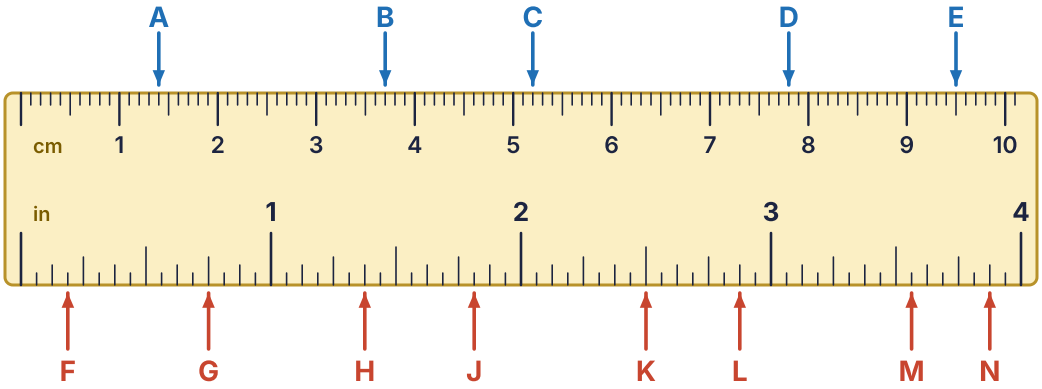


Figure 6.3 Rule for Practice 6.1.

PRACTICE 6.1 **Reading a rule**

Use Figure 6.3. Give A to E in millimetres. Give F to N in inches, in lowest terms.

1. Point A
2. Point B
3. Point C
4. Point D
5. Point E
6. Point F
7. Point G
8. Point H
9. Point J
10. Point K
11. Point L
12. Point M
13. Point N
14. Write the reading for point C in centimetres.
15. How many millimetres is it from point A to point D?
16. How far is it from point G to point K, in inches?

6.3 Reading gauges and meters

A dial is a ruler bent into an arc. The same question unlocks every one of them: **what is each small space worth?**

KEY RULE **Finding the scale value**

1. Pick two neighbouring **numbered** marks and subtract them.

2. Count the **spaces** between them (spaces, not marks).

3. Divide: value of one space = difference ÷ number of spaces.

4. Read the needle: last numbered mark + spaces past it × value of a space. If the needle is between two marks, estimate.

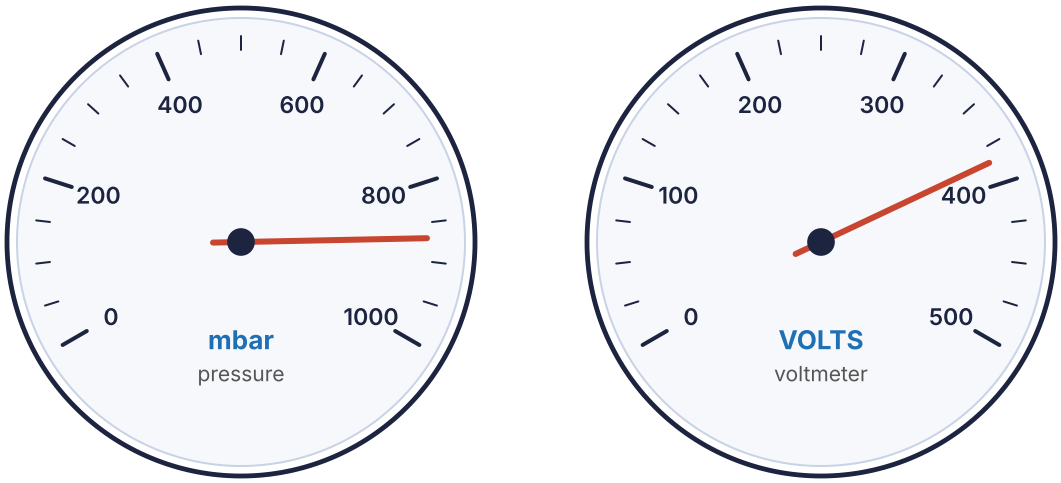


Figure 6.4 Two dials. Left: numbered every 200 with 4 spaces between numbers. Right: numbered every 100 with 4 spaces.

EXAMPLE 6.3

Estimate the reading of the pressure gauge on the left of Figure 6.4.

200 − 0 = 200	Two neighbouring numbers differ by 200.
200 ÷ 4 spaces = 50 per space	There are four spaces between numbered marks.
800 + 1 space = 850	The needle is past 800 and past the first small mark.
850 + a little less than half a space ≈ 870	It sits not quite halfway to 900. Half a space would be 25, so estimate about 20.

ANSWER **about 870 mbar**

On the voltmeter at the right, each space is 100 ÷ 4 = 25 volts. The needle is three spaces past 300 (that is 375) and a bit more, so the reading is about 385 volts. An estimate between marks will differ slightly from person to person; 385 and 390 are both reasonable, 350 is not.

WATCH OUT **Count spaces, and check your scale value**

If you think each space is worth 20, count by 20s from one number to the next. If you do not land exactly on the next number, the scale value is wrong. On the gauge above: 0, 50, 100, 150, 200. It lands, so 50 is right.

Gauges with two scales

Many pressure gauges carry two scales on one face, for example **kPa** on the outer ring and **psi** on the inner ring. There is one needle but two readings, and each scale has its **own** numbers and its own scale value.

- Decide which scale the question asks for, and look only at that ring of numbers.
- Work out the scale value for that ring separately. The outer ring might be worth 1000 per space while the inner is worth 100.
- Follow the needle to the ring you are reading. Do not read the psi position against the kPa numbers.
- As a check, 1 psi is about 6.9 kPa, so the kPa reading should be roughly 7 times the psi reading.

EXAMPLE 6.4

A dual-scale gauge reads 6000 kPa on its outer scale. Use the check above to estimate what the inner psi scale should read.

6000 ÷ 6.9 = 869.6	kPa is the smaller unit, so the psi number is smaller: divide.
≈ 870	A reading between 860 and 880 psi is consistent.

ANSWER **about 870 psi**

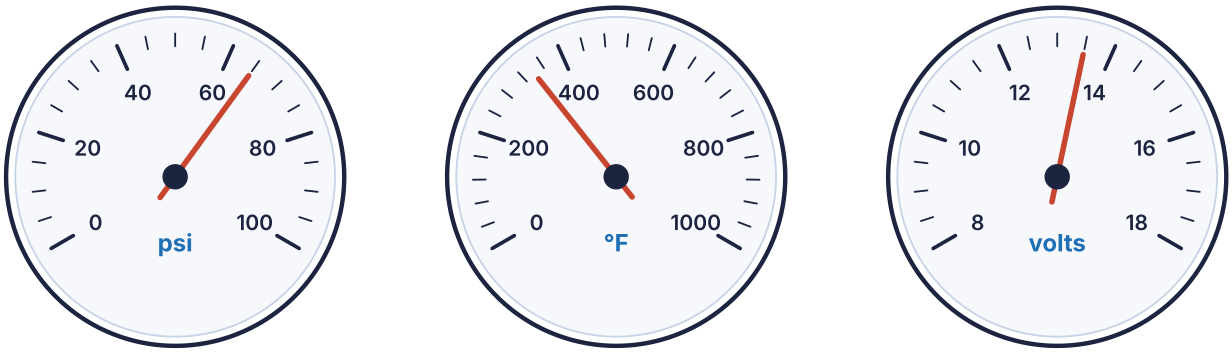


Figure 6.5 Gauges A, B and C for Practice 6.2.

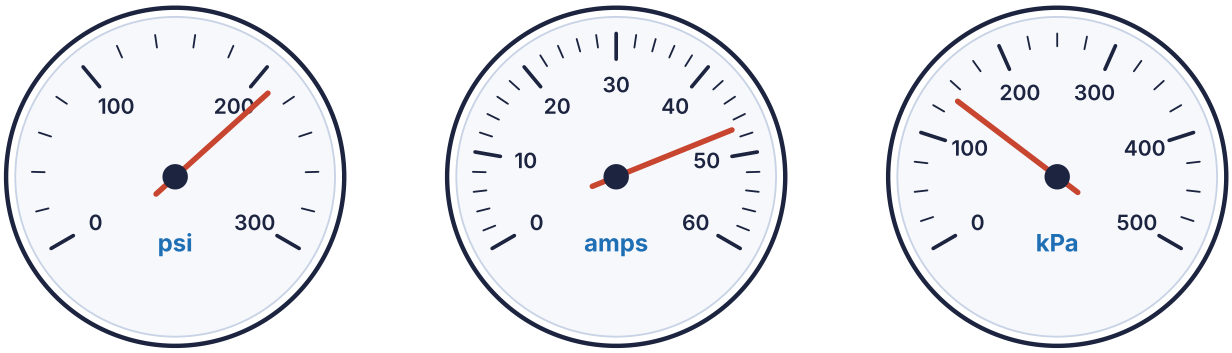


Figure 6.6 Gauges D, E and F for Practice 6.2.

PRACTICE 6.2 **Reading gauges**

Use Figures 6.5 and 6.6. Estimates within half a space of the answer are acceptable.

1. Gauge A: what is one small space worth?
2. Gauge A: estimate the reading.
3. Gauge B: what is one small space worth?
4. Gauge B: estimate the reading.
5. Gauge C: what is one small space worth?
6. Gauge C: estimate the reading.
7. Gauge D: what is one small space worth?
8. Gauge D: estimate the reading.
9. Gauge E: what is one small space worth?
10. Gauge E: estimate the reading.
11. Gauge F: what is one small space worth?
12. Gauge F: estimate the reading.

6.4 Tolerance

Nothing can be made to an exact size. A drawing therefore gives a **nominal** size and a **tolerance**, the amount the part may be over or under. The sign ± is read "plus or minus."

KEY RULE **Limits of a dimension**

upper limit = nominal size + tolerance

lower limit = nominal size − tolerance

A part is acceptable if its size is anywhere from the lower limit to the upper limit, including the limits themselves.

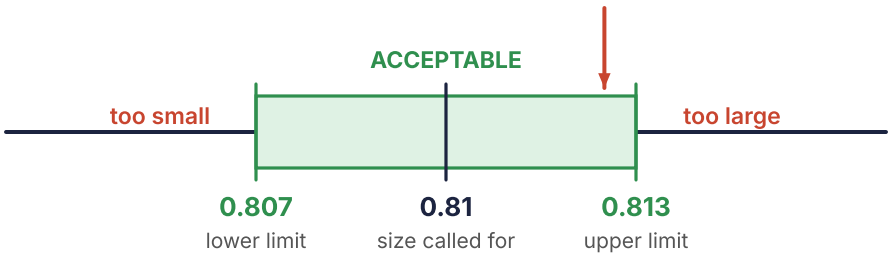


Figure 6.7 The tolerance band for 0.81 ± 0.003 in. The arrow marks a 13/16-in hole at 0.8125 in.

EXAMPLE 6.5

A hole is specified as 0.81 in ± 0.003 in. (a) What are the limits? (b) Will a hole drilled with a $\frac{13}{16}$-in bit be within specification?

0.810 + 0.003 = 0.813	(a) Upper limit. Write 0.81 as 0.810 so the decimal places line up.
0.810 − 0.003 = 0.807	Lower limit.
13 ÷ 16 = 0.8125	(b) Change the drill size to a decimal so it can be compared.
0.807 ≤ 0.8125 ≤ 0.813	Compare using four decimal places: 0.8070, 0.8125, 0.8130. It lies between the limits.

ANSWER **(a) 0.807 in to 0.813 in (b) Yes**

WATCH OUT **Line up the decimal places before adding**

0.81 + 0.003 is not 0.84 and not 0.813 by luck. Write 0.810 + 0.003. A tolerance usually has more decimal places than the nominal size.

Fraction and percent tolerances

Tolerances on a tape-measure job are fractions: $24\ \frac{1}{2}$ in $\pm\ \frac{1}{16}$ in gives limits of $24\ \frac{7}{16}$ in and $24\ \frac{9}{16}$ in. Electrical parts are often toleranced by percent: a 200-ohm resistor at ±5% may be off by 5% of 200 = 10 ohms, so its limits are 190 and 210 ohms.

6.5 Finding a missing dimension

Drawings rarely repeat information. If the overall length and all but one of the parts are given, the missing part is found by subtracting. The idea is simply: **the parts add up to the whole.**

KEY RULE **Missing length**

missing part = overall length − (sum of the known parts)

Change every dimension to the same form first: all decimals or all fractions.

EXAMPLE 6.6

A bar is 78 $\frac{3}{4}$ in long overall. It is marked in three sections. The first is 11.25 in and the third is 31 $\frac{3}{8}$ in. How long is the middle section? Do not round.

78 $\frac{3}{4}$ = 78.75; 31 $\frac{3}{8}$ = 31.375	Change the fractions to decimals.
78.750 − 11.250 − 31.375	Subtract each known part from the overall length.
= 36.125	No rounding is needed or allowed.
0.125 = $\frac{1}{8}$	As a mixed number, if wanted.

ANSWER **36.125 in, or 36 $\frac{1}{8}$ in**

PRACTICE 6.3 **Tolerance and missing lengths**

Do not round.

1. 2.50 in ± 0.01 in. Find the limits. Is a part measuring 2.495 in acceptable?
2. 0.375 in ± 0.005 in. Find the limits. Is a part measuring 0.382 in acceptable?
3. 1.25 in ± 0.002 in. Find the limits. Is a part measuring 1.2485 in acceptable?
4. 4.0 in ± 0.05 in. Find the limits. Is a part measuring 3.94 in acceptable?
5. 0.62 in ± 0.003 in. Find the limits. Is a part measuring $\frac{5}{8}$ in acceptable?
6. 0.44 in ± 0.004 in. Find the limits. Is a part measuring $\frac{7}{16}$ in acceptable?
7. Overall length 60 in; known parts 14.5 in, 22 $\frac{3}{8}$ in. Find the missing part.
8. Overall length 96 $\frac{1}{2}$ in; known parts 18.25 in, 40 $\frac{5}{8}$ in. Find the missing part.
9. Overall length 42 $\frac{3}{8}$ in; known parts 9.75 in, 12 $\frac{1}{2}$ in, 6.125 in. Find the missing part.
10. Overall length 120 in; known parts 33 $\frac{7}{8}$ in, 33 $\frac{7}{8}$ in, 20.5 in. Find the missing part.
11. A 150-Ω resistor has a tolerance of ±10%. What are its limits?
12. A stud is to be cut 92 $\frac{5}{8}$ in ± $\frac{1}{8}$ in. What are the limits?

CHAPTER SUMMARY

- Metric rule: the numbers are centimetres and each small space is 1 mm.
- Inch rule: count sixteenths past the last whole inch, then reduce the fraction.
- Gauge: scale value = difference between numbered marks ÷ number of spaces between them.
- On a dual-scale gauge, read each scale with its own numbers and its own scale value.
- Limits = nominal ± tolerance. A part on or between the limits is acceptable.
- A missing dimension is the overall length minus the known parts.

CHAPTER 6 REVIEW

1. How many millimetres are in 7.3 cm?
2. A point is 4 small marks past the 6 on a centimetre scale. Give the reading in mm.
3. On an inch rule in sixteenths, a point is 12 spaces past the 2. Give the reading in lowest terms.
4. A point is 14 sixteenths past the 5-inch mark. Give the reading in lowest terms.
5. Which mark is longer on an inch rule, $\frac{3}{8}$ or $\frac{3}{4}$?
6. A gauge is numbered 0, 40, 80, 120 with 8 spaces between numbers. What is each space worth?
7. On that gauge the needle is 3 spaces past 80. What is the reading?
8. A gauge is numbered every 500 with 5 spaces between numbers. The needle is halfway between the 2nd and 3rd marks past 1500. Estimate the reading.
9. A dual-scale gauge reads 100 psi. About what should the kPa scale read?
10. Find the limits of 1.75 in ± 0.015 in.
11. A shaft must be 0.50 in ± 0.002 in. Is a shaft measuring 0.5025 in acceptable?
12. A hole must be 0.69 in ± 0.005 in. Is an $\frac{11}{16}$-in hole acceptable?
13. Overall length 54 $\frac{1}{4}$ in; known parts 20.5 in and 16 $\frac{3}{8}$ in. Find the missing part without rounding.
14. Four equal spaces and two end margins of 1.25 in make up a 30-in panel. How wide is each space?

APPLIED TECHNICAL MATH

Part 3 · Algebra

A formula is a set of instructions written in symbols. Part 3 covers the order in which those instructions are carried out, how to put numbers into a formula, and how to run a formula backward to find a missing value.

CHAPTER 7

Order of Operations

Ask two people to work out 6 + 4 ÷ 2 and you may get 5 from one and 8 from the other. Only one of them is right. Mathematics settles the question with a fixed order of operations, and every formula in the trades depends on it.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- State the order of operations and explain why an agreed order is needed.
- Recognise every kind of grouping symbol, including the fraction bar.
- Evaluate expressions with exponents and square roots.
- Work multiplication with division, and addition with subtraction, from left to right.
- Show the work for a multi-step evaluation one operation at a time.

KEY TERMS

expression	Numbers and operations written together, such as 6 + 4 ÷ 2.
evaluate	To work an expression down to a single number.
exponent	The small raised number that tells how many times the base is multiplied by itself: $4^2 = 4 \times 4$.
square root	The number that, multiplied by itself, gives the number under the √ sign: $\sqrt{25} = 5$.
grouping symbol	Anything that says “do this part first”: parentheses, brackets, braces, absolute-value bars, a fraction bar, a root sign.

7.1 Why an order is needed

Consider 6 + 4 ÷ 2. Working from left to right gives 10 ÷ 2 = 5. Dividing first gives 6 + 2 = 8. An expression must have one value, so everyone agrees to the same order. Under that agreement division comes before addition, and the answer is **8**.

> Wrong: 6 + 4 ÷ 2 = 10 ÷ 2 = 5
>
> Right: 6 + 4 ÷ 2 = 6 + 2 = 8



Figure 7.1 The order of operations. Four levels, worked from the top down.

KEY RULE **The order of operations (PEMDAS)**

1. P: Parentheses and every other grouping symbol: ( ), [ ], { }, absolute-value bars, and the top and bottom of a fraction bar.

2. E: Exponents and roots.

3. M D: Multiplication and division, whichever comes first, working from **left to right**.

4. A S: Addition and subtraction, whichever comes first, working from **left to right**.

WATCH OUT **PEMDAS has four steps, not six**

Multiplication does **not** come before division, and addition does **not** come before subtraction. Each pair shares a level and is worked in the order it appears, left to right. $24 \div 4 \times 2$ is $6 \times 2 = 12$, not $24 \div 8 = 3$.

7.2 Exponents and roots

An **exponent** is shorthand for repeated multiplication. In 4^2, the 4 is the **base** and the 2 is the exponent: use 4 as a factor twice. A **square root** undoes a square.

> $4^2 = 4 \times 4 = 16$ $2^3 = 2 \times 2 \times 2 = 8$ $6^4 = 6 \times 6 \times 6 \times 6 = 1296$
>
> $\sqrt{16} = 4$ because $4 \times 4 = 16$ $\sqrt{2.25} = 1.5$ because $1.5 \times 1.5 = 2.25$

WATCH OUT **An exponent is not multiplication by the exponent**

4^2 is $4 \times 4 = 16$, not $4 \times 2 = 8$. And 2^3 is 8, not 6.

A root sign is also a grouping symbol. Everything under the bar must be worked out **before** the root is taken: $\sqrt{(9 + 16)}$ is $\sqrt{25} = 5$. It is not $3 + 4$.

CALCULATOR TIP **Powers and roots on a calculator**

Use the `x²` key for squares and the `^` or `xʸ` key for other powers: 6^4 is `6 ^ 4`. For a root, press `√` and then put the whole quantity in parentheses: `√(9 + 16)`. On some calculators the root key is the shifted function of `x²`.

7.3 Multiplying without a times sign

In algebra the × sign is usually left out, because it looks like the letter x. A number written directly against a parenthesis, or against a letter, means multiply.

> 3(8) means 3 × 8 2(5 + 1) means 2 × (5 + 1) 4C means 4 × C (3)(7) means 3 × 7

CALCULATOR TIP **Some calculators need the times sign**

Many calculators accept 3(8) as typed. Others report a syntax error or, worse, read it as something else. If yours does, key in the × yourself: `3 × (8)`.

7.4 Working an expression step by step

Do one operation per line and copy everything else down unchanged. It feels slow, but it is how errors are found, and it is the work an instructor expects to see.

EXAMPLE 7.1

Evaluate $5 - 3(4^2 \div 2) + 30$.

$5 - 3(\mathbf{16} \div 2) + 30$	Start inside the parentheses. Inside, the exponent comes first: $4^2 = 16$.
$5 - 3(\mathbf{8}) + 30$	Still inside the parentheses: $16 \div 2 = 8$.
$5 - \mathbf{24} + 30$	The parentheses now hold one number. Multiply: $3(8) = 24$.
$\mathbf{-19} + 30$	Add and subtract from left to right: $5 - 24 = -19$.
11	$-19 + 30 = 11$.

ANSWER **11**

WATCH OUT **Subtraction before addition, when it comes first**

In 5 − 24 + 30, do **not** add 24 + 30 first. That gives 5 − 54 = −49, which is wrong. Work left to right: 5 − 24, then + 30.

EXAMPLE 7.2

Evaluate $18 - 12 \div 3 \times 2 + 5^2$.

$18 - 12 \div 3 \times 2 + \mathbf{25}$	No grouping symbols. Exponent first.
$18 - \mathbf{4} \times 2 + 25$	Multiplication and division left to right: the division comes first, $12 \div 3 = 4$.
$18 - \mathbf{8} + 25$	Then $4 \times 2 = 8$.
$\mathbf{10} + 25$	Left to right: $18 - 8 = 10$.
35	

ANSWER **35**

The fraction bar

A long fraction bar groups everything above it and everything below it. Work the entire top to one number, work the entire bottom to one number, and divide last.

EXAMPLE 7.3

Evaluate $\frac{12 - 3(4^2 \div 2^3) + 9}{5^2 - 10}$. Round to the nearest hundredth if needed.

top: $12 - 3(16 \div 8) + 9$	Inside the parentheses, exponents first: $4^2 = 16$ and $2^3 = 8$.
top: $12 - 3(2) + 9 = 12 - 6 + 9 = 15$	Finish the parentheses, multiply, then go left to right.
bottom: $25 - 10 = 15$	Exponent, then subtract.
$\frac{15}{15} = 1$	Only now divide the top by the bottom.

ANSWER **1**

CALCULATOR TIP **Entering a fraction bar**

A calculator does not see a fraction bar unless you tell it. Put the whole top in parentheses and the whole bottom in parentheses: `(12 − 3 × (4² ÷ 2³) + 9) ÷ (5² − 10)`. Without them the calculator divides only the 9 by the 25.

Grouping inside grouping

When grouping symbols are nested, begin with the innermost pair and work outward.

EXAMPLE 7.4

Evaluate $2[7 + 3(10 - 6)]^2$.

$2[7 + 3(\mathbf{4})]^2$	Innermost parentheses first.
$2[7 + \mathbf{12}]^2$	Inside the brackets, multiply before adding.
$2[\mathbf{19}]^2$	Finish the brackets.
$2 \times \mathbf{361}$	Exponent before multiplication: $19^2 = 361$.
722	

ANSWER **722**

Negative numbers

The Chapter 7 work occasionally passes through a negative number, as −19 did above. Two facts are enough: subtracting a larger number from a smaller one gives a negative result ($5 - 24 = -19$), and a negative times a positive is negative ($(19)(-5) = -95$).

PRACTICE 7.1 **Order of operations**

Evaluate. Show one operation per line. Round to the nearest hundredth only if needed.

1. $6 + 4 \div 2$
2. $20 - 8 + 5$
3. $24 \div 4 \times 2$
4. $3 + 5 \times 2^2$
5. $(3 + 5) \times 2^2$
6. $40 \div (2 + 3) \times 4$
7. $7 - 2(3^2 - 4) + 15$
8. $5 + 12 \div 4 \times 15 - 3^2$
9. $\sqrt{(36 + 64)}$
10. $4 + \sqrt{81} \times 2$
11. $\dfrac{18 + 6}{2 \times 3}$
12. $\dfrac{5^2 - 1}{4 + 2^2}$
13. $\dfrac{5 + 12 \div 4 \times 15 + 15^2}{6}$
14. $3[20 - 2(4 + 3)]$
15. $100 - 4(2 + 3)^2$
16. $2^3 + 6(9 - 5) \div 8$
17. $(7 + 12)(7 - 12)$
18. $\dfrac{8(6 + 14)}{8 + 6 + 14}$

CHAPTER SUMMARY

- Order: grouping symbols, then exponents and roots, then × and ÷ left to right, then + and − left to right.
- A fraction bar and a root sign are grouping symbols.
- A number against a parenthesis means multiply.
- Write one operation per line and copy the rest unchanged.
- On a calculator, a fraction needs parentheses around its top and around its bottom.

CHAPTER 7 REVIEW

1. In your own words, list the four levels of the order of operations.
2. $15 - 6 + 4$
3. $36 \div 6 \times 3$
4. $9 + 3 \times 4^2$
5. $(9 + 3) \times 4^2$
6. $50 - 2(3 + 4^2)$
7. $8 - 5(6^2 \div 9) + 20$
8. $\sqrt{(5^2 + 12^2)}$
9. $3\sqrt{49} - 2^3$
10. $\dfrac{30 - 2(3^2 - 4)}{2^2 + 1}$
11. $\dfrac{14 - 2(5^2 \div 5) + 8}{3^2 - 5}$
12. $4[2 + 3(8 - 6)^2]$
13. $\dfrac{6^2 + 8^2}{2 \times 5}$
14. $2.5(4.2 - 1.7)^2$ (do not round)
15. $\dfrac{16(6 + 8)}{16 + 6 + 8}$ (nearest hundredth)
16. A student evaluates $30 - 10 + 5$ and gets 15. What was the mistake, and what is the answer?
17. Write the keystrokes, with parentheses, for $\dfrac{7 + 9}{2 \times 4}$.
18. Does $3(4)^2$ equal 144 or 48? Explain.

CHAPTER 8

Evaluating Formulas

The trades run on formulas: for the load a post will carry, the speed of a grinding wheel, the capacitance of a circuit. Nobody memorises them all. What matters is the skill of putting the right numbers in and working them out correctly.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Explain what a formula is and what its letters stand for.
- Substitute given values into a formula, showing the substitution.
- Evaluate the result using the order of operations.
- Evaluate formulas containing exponents, roots and fraction bars.
- Round and label the answer as a problem directs.

KEY TERMS

formula	A rule written with letters that tells how to calculate one quantity from others.
variable	A letter that stands for a number.
substitute	To replace each letter with its given number.
subscript	A small lowered label that tells similar letters apart, as in C_1 and C_2.
constant	A number in a formula that never changes, such as the 12 in $L = \pi dr \div 12$.

8.1 What a formula says

A **formula** is a sentence in shorthand. The formula $A = LW$ says "the area of a rectangle equals its length times its width." Each letter is a **variable**: a placeholder waiting for a number. Letters written side by side are multiplied.

- Capital and lowercase letters are **different** variables. In $N = \sqrt{(D \div d)^5}$, D and d are two different diameters.
- A **subscript** makes a new variable name. C_1, C_2 and C_3 are three different capacitors. The subscript is a label, not a number to calculate with.
- Numbers that appear in a formula, such as 1.085 or 12, are **constants**. Leave them exactly as they are.

8.2 Substitute, then evaluate

KEY RULE **Three lines for every formula problem**

1. Write the formula.

2. Substitute. Rewrite the formula with each letter replaced by its number **in parentheses**. Change nothing else.

3. Evaluate by the order of operations, then round and write the unit.

The parentheses in step 2 matter. They keep a multiplication from disappearing (2*C* with $C = 30$ must become 2(30), not 230) and they keep negative numbers and fractions together.

EXAMPLE 8.1

Evaluate L = 3A + 4(B + C) for A = 2, B = 3 and C = 4.

L = 3A + 4(B + C)	Write the formula.
L = 3(2) + 4(3 + 4)	Substitute. Every letter is replaced; nothing else moves.
L = 3(2) + 4(7)	Parentheses first.
L = 6 + 28	Multiply.
L = 34	Add.

ANSWER **L = 34**

EXAMPLE 8.2 Two variables that look alike

Evaluate $2C + 1.6(D + d) + \frac{D + d}{4C}$ for C = 30 in, D = 10 in and d = 5 in. Give a mixed number or a non-rounded decimal.

$2(30) + 1.6(10 + 5) + \frac{10 + 5}{4(30)}$	Substitute.
$2(30) + 1.6(15) + \frac{15}{120}$	Grouping: the parentheses, and the top and bottom of the fraction.
60 + 24 + 0.125	Multiply and divide.
84.125	Add. The problem forbids rounding, so every digit stays.

ANSWER **84.125 in, or $84\ \frac{1}{8}$ in**

WATCH OUT **Show the substitution line**

Going straight from the formula to the answer hides where an error happened and usually costs credit. The substitution line is the proof that the right number went in the right place.

8.3 Formulas from the trades

The examples that follow use real formulas. In each one, read the description to match every letter with its number, and note how the answer is to be rounded.

EXAMPLE 8.3

Machine trades. The linear speed of a grinding wheel in feet per minute is $L = \frac{\pi dr}{12}$, where d is the wheel diameter in inches and r is the speed in rpm. Find the linear speed of an 8-in wheel turning at 1750 rpm. Use $\pi = 3.14$ and round to the nearest whole number.

$L = \frac{\pi dr}{12}$	Formula.
$L = \frac{(3.14)(8)(1750)}{12}$	Substitute.
$L = \frac{43{,}960}{12}$	Multiply across the top.
$L = 3663.33\ldots$	Divide.
$L = 3663$	Round as directed.

ANSWER **3663 ft/min**

EXAMPLE 8.4 Exponents

Construction. The crushing load in tons of a square wooden post is $L = \frac{25T^4}{H^2}$, where T is the thickness in inches and H is the height in feet. Find the crushing load of a 4-in post 8 ft high.

$L = \frac{25T^4}{H^2}$	Formula.
$L = \frac{25(4)^4}{(8)^2}$	Substitute.
$L = \frac{25(256)}{64}$	Exponents first: $4^4 = 256$ and $8^2 = 64$.
$L = \frac{6400}{64} = 100$	Multiply the top, then divide.

ANSWER **100 tons**

WATCH OUT **The exponent belongs only to what it touches**

In $25T^4$ the exponent applies to T alone. Raise 4 to the fourth power **first**, then multiply by 25. Multiplying 25 × 4 first and then raising 100 to the fourth power gives one hundred million.

EXAMPLE 8.5 A root

Plumbing. To carry the same flow as one pipe of diameter D, the number of smaller pipes of diameter d needed is $N = \sqrt{(D \div d)^5}$. How many 2-in pipes are needed to equal one 6-in pipe? Round to the nearest whole number.

$N = \sqrt{\left(\frac{D}{d}\right)^5}$	Formula.
$N = \sqrt{\left(\frac{6}{2}\right)^5}$	Substitute.
$N = \sqrt{(3)^5}$	Innermost grouping first: 6 ÷ 2 = 3.
$N = \sqrt{243}$	The exponent is under the root sign, so it is done before the root: $3^5 = 243$.
$N = 15.588\ldots$	Take the square root.
$N = 16$	Round to the nearest whole number.

ANSWER **16 pipes**

CALCULATOR TIP **Keying a root of a power**

Enter it the way it is grouped: `√( (6 ÷ 2) ^ 5 )`. Open a parenthesis after the root sign and close it at the very end, so the root covers the whole power.

EXAMPLE 8.6 Subscripts and a fraction bar

Electronics. For one capacitor in series with two in parallel, the total capacitance is $C_T = \frac{C_1(C_2 + C_3)}{C_1 + C_2 + C_3}$. Find C_T when $C_1 = 12\ \mu F$, $C_2 = 5\ \mu F$ and $C_3 = 7\ \mu F$. Round to the nearest hundredth.

$C_T = \frac{12(5 + 7)}{12 + 5 + 7}$	Substitute. The subscripts only say which capacitor is which.
$C_T = \frac{12(12)}{24}$	Parentheses on top; add across the bottom.
$C_T = \frac{144}{24} = 6$	Multiply the top, then divide.

ANSWER **6 µF**

EXAMPLE 8.7

Manufacturing. The capacity of a feeder belt in metric tons per hour is $C = \frac{W^2DS}{1.085}$, where W is the belt width in metres, D is the density of the material in kg/m³ and S is the belt speed in m/s. Find the capacity for W = 1.2 m, D = 800 kg/m³ and S = 2.5 m/s. Round to the nearest whole number.

$C = \frac{(1.2)^2(800)(2.5)}{1.085}$	Substitute.
$C = \frac{(1.44)(800)(2.5)}{1.085}$	Exponent first.
$C = \frac{2880}{1.085}$	Multiply across the top.
$C = 2654.37\ldots$	Divide.
$C = 2654$	Round.

ANSWER **2654 metric tons per hour**

PRACTICE 8.1 **Substitute and evaluate**

Write the formula, show the substitution, and evaluate. Do not round unless told to.

1. $N = (a + b)(a - b)$; $a = 9, b = 14$
2. $L = \frac{s(P + p)}{2}$; $s = 4.5, P = 42, p = 30$
3. $f = \frac{1}{8N}$; $N = 5$ (answer as a fraction)
4. $t = \frac{D - d}{L}$; $D = 15, d = 6, L = 3$
5. $V = \frac{gt^2}{2}$; $g = 32.2, t = 2.5$
6. $P = 2L + 2W$; $L = 18.5$ ft, $W = 12.25$ ft
7. $A = \frac{(b_1 + b_2)h}{2}$; $b_1 = 9, b_2 = 14, h = 6$
8. $c = \sqrt{(a^2 + b^2)}$; $a = 9, b = 12$
9. **Welding.** Tensile strength $T = \frac{L}{A}$ psi. $L = 9{,}800$ lb, $A = 0.14$ sq in.
10. **Electrical.** Power $P = I^2R$ watts. $I = 2.5$ A, $R = 48\ \Omega$.
11. **Machine trades.** $L = \frac{\pi dr}{12}$. $d = 10$ in, $r = 1200$ rpm, $\pi = 3.14$.
12. **Construction.** $L = \frac{25T^4}{H^2}$. $T = 6$ in, $H = 10$ ft.
13. **Plumbing.** $N = \sqrt{(D \div d)^5}$. $D = 4$ in, $d = 1$ in.
14. **Plumbing.** $N = \sqrt{(D \div d)^5}$. $D = 3$ in, $d = \frac{3}{4}$ in. Round to the nearest whole number.
15. **Electronics.** Two inductors in parallel: $L_T = \frac{L_1L_2}{L_1 + L_2}$. $L_1 = 300$ mH, $L_2 = 60$ mH.

16. **Electronics.** $C_T = \frac{C_1(C_2 + C_3)}{C_1 + C_2 + C_3}$. $C_1 = 20$, $C_2 = 4$, $C_3 = 9$ (μF). Nearest hundredth.

17. **Electronics.** Inductance of a coil $L = \frac{R^2N^2}{9R + 10D}$ μH. R = 2 in, D = 5 in, N = 150 turns. Nearest whole number.

18. **Automotive.** Indicated horsepower $ihp = \frac{MEP \times D \times S}{792{,}000}$. MEP = 150 psi, D = 300 cu in, S = 3300 rpm.

19. **Hydraulics.** Force on a piston $F = \frac{\pi PD^2}{4}$ lb. P = 40 psi, D = 2.5 in, π = 3.14. Nearest whole number.

20. **Manufacturing.** $C = \frac{W^2DS}{1.085}$. W = 0.9 m, D = 1500 kg/m³, S = 2 m/s. Nearest whole number.

CHAPTER SUMMARY

- A formula is a rule; its letters are placeholders for numbers.
- Write the formula, substitute with parentheses, then evaluate by the order of operations.
- Capital and lowercase letters, and letters with different subscripts, are different variables.
- An exponent applies only to the quantity it touches.
- Round only at the end, as the problem directs, and write the unit.

CHAPTER 8 REVIEW

1. Evaluate $L = 5A + 2(B + C)$ for A = 3, B = 6, C = 2.
2. Evaluate $3C + 2.4(D - d)$ for C = 12, D = 9, d = 4.
3. Evaluate $\frac{D + d}{2C}$ for D = 14, d = 6, C = 8. Give a decimal.
4. Evaluate $N = (a + b)(a - b)$ for a = 5, b = 11.
5. Evaluate $V = \frac{gt^2}{2}$ for g = 32.2 and t = 3.
6. Evaluate $A = \sqrt{(s(s - a)(s - b)(s - c))}$ for s = 12, a = 6, b = 8, c = 10.
7. Tensile strength $T = \frac{L}{A}$. L = 15,000 lb, A = 0.21 sq in. Round to the nearest hundred psi.
8. Grinding wheel $L = \frac{\pi dr}{12}$. d = 7 in, r = 3000 rpm, π = 3.14.
9. Crushing load $L = \frac{25T^4}{H^2}$. T = 8 in, H = 16 ft.
10. Pipes $N = \sqrt{(D \div d)^5}$. D = 12 in, d = 4 in. Nearest whole number.
11. Feeder belt $C = \frac{W^2DS}{1.085}$. W = 1.5 m, D = 600, S = 3. Nearest whole number.
12. Piston force $F = \frac{\pi PD^2}{4}$. P = 30 psi, D = 4 in, π = 3.14.
13. $C_T = \frac{C_1(C_2 + C_3)}{C_1 + C_2 + C_3}$ for $C_1 = 18$, $C_2 = 7$, $C_3 = 5$. Nearest hundredth.
14. $L_T = \frac{L_1L_2}{L_1 + L_2}$ for $L_1 = 120$ mH and $L_2 = 80$ mH.
15. In the formula $C = \frac{W^2DS}{1.085}$, which symbols are variables and which is a constant?
16. A student substitutes C = 30 into 2C and writes 230. What should have been written?

CHAPTER 9

Solving Formulas

In Chapter 8 the unknown always stood alone on the left: put the numbers in and calculate. On the job it is often the other way round. You know the volume a tank must hold and need its height; you know the perimeter of a fence and need its length. Then the formula has to be run backward.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

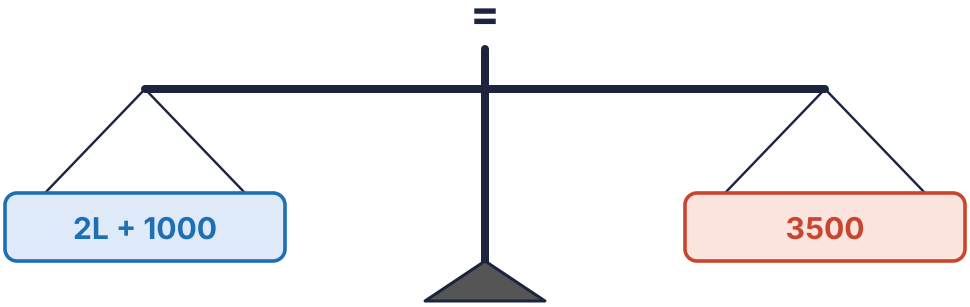
- Explain what it means to solve an equation.
- Undo an operation with its inverse, keeping the equation balanced.
- Solve one-step and two-step equations.
- Substitute into a formula and solve for the unknown that is left.
- Solve equations that contain fractions without rounding.

KEY TERMS

equation	A statement that two expressions are equal.
solve	To find the value of the unknown that makes the equation true.
inverse operation	The operation that undoes another: + and −, × and ÷, square and square root.
isolate	To get the unknown alone on one side of the equals sign.
coefficient	The number multiplying a variable: the 2 in 2L.

9.1 Equations and balance

An equation is like a balance scale: the two sides weigh the same. Whatever is done to one side must be done to the other, or the scale tips and the equation is no longer true.



Take 1000 from both pans and the scale stays level: 2L = 2500.

Figure 9.1 An equation is a balance.

KEY RULE **The golden rule of equations**

Do the same thing to both sides.

You may add the same number to both sides, subtract the same number from both sides, multiply both sides by the same number, or divide both sides by the same number (except zero).

To **solve** an equation is to get the unknown letter alone: x = a number. That is done by removing everything else from its side, one operation at a time, using **inverse operations**.

If the unknown has had this done to it	Undo it by	Example
a number added: $x + 7 = 19$	subtracting it from both sides	$x = 19 - 7 = 12$
a number subtracted: $x - 7 = 19$	adding it to both sides	$x = 19 + 7 = 26$
multiplied by a number: $7x = 21$	dividing both sides by it	$x = 21 \div 7 = 3$
divided by a number: $\frac{x}{7} = 21$	multiplying both sides by it	$x = 21 \times 7 = 147$

9.2 One-step equations

EXAMPLE 9.1

Ohm's law is $I = \frac{E}{R}$. A circuit carries a current I = 0.60 A through a resistance R = 400 Ω. Find the voltage E.

$I = \frac{E}{R}$	Formula.
$0.60 = \frac{E}{400}$	Substitute what is known. One letter is left.
$0.60 \times 400 = \frac{E}{400} \times 400$	E is divided by 400. Undo it: multiply both sides by 400.
$240 = E$	The 400s cancel on the right.

ANSWER **E = 240 V**

EXAMPLE 9.2

A semi-trailer's volume is V = LWH. It holds V = 3600 cu ft and is L = 50 ft long and W = 8.5 ft wide. Find its height to the nearest tenth.

$3600 = (50)(8.5)H$	Substitute.
$3600 = 425H$	Simplify the right side first: 50 × 8.5 = 425.
$\frac{3600}{425} = \frac{425H}{425}$	H is multiplied by 425. Divide both sides by 425.
$8.4705\ldots = H$	
$H = 8.5$	Round to the nearest tenth.

ANSWER **H = 8.5 ft**

KEY RULE **Simplify before you solve**

After substituting, combine everything that can be combined on each side. Then start undoing. In the trailer example, multiplying 50 × 8.5 first left a simple one-step equation.

It makes no difference which side the unknown ends up on: 240 = *E* says the same thing as *E* = 240.

PRACTICE 9.1 **One-step equations**

Substitute and solve. Show the step that undoes the operation.

1. **Aviation.** D = RT. D = 1,400 mi, R = 250 mph. Find T.
2. **Sheet metal.** Water pressure P = 62.4D lb/sq ft at depth D ft. P = 936. Find D.
3. **Physics.** fw = 343. f = 250. Find w.
4. **Physics.** PV = 1080. V = 45 cu ft. Find P.
5. **Electrical.** Transformer E = 40W. E = 1,000 V. Find W.
6. **Electrical.** $I = \frac{E}{R}$. I = 1.5 A, R = 80 Ω. Find E.
7. **Welding.** $T = \frac{L}{A}$. T = 60,000 psi, A = 0.25 sq in. Find L.
8. A = LW. A = 187 sq ft, W = 11 ft. Find L.
9. V = LWH. V = 2,880 cu ft, L = 40 ft, H = 9 ft. Find W.
10. $x + 14.5 = 32$
11. $y - 3.75 = 12$
12. $\frac{n}{6} = 4.5$

9.3 Two-step equations

When the unknown has been multiplied **and** something has been added or subtracted, there are two operations to undo. Undo them in the **reverse** of the order of operations: first the addition or subtraction, then the multiplication or division. Think of taking off shoes and socks: the last thing put on is the first thing taken off.

KEY RULE **Solving a two-step equation**

1. Substitute and simplify each side.

2. Undo the **addition or subtraction**: move the plain number away from the unknown's side.

3. Undo the **multiplication or division**: divide by the coefficient.

4. Check by putting the answer back into the original equation.

EXAMPLE 9.3

The perimeter of a rectangle is P = 2L + 2W. A pasture is to be fenced with P = 3200 ft of fence and is W = 450 ft wide. Find its length.

$3200 = 2L + 2(450)$	Substitute.
$3200 = 2L + 900$	Simplify: 2(450) = 900.
$3200 - 900 = 2L + 900 - 900$	900 is added to the 2L. Subtract 900 from both sides.
$2300 = 2L$	
$\frac{2300}{2} = \frac{2L}{2}$	L is multiplied by 2. Divide both sides by 2.
$1150 = L$	
Check: $2(1150) + 2(450) = 2300 + 900 = 3200$ ✓	The answer makes the original equation true.

ANSWER **L = 1150 ft**

EXAMPLE 9.4

A service call is billed as T = 45 + 60H dollars, where H is the number of hours. A bill came to T = $345. How many hours were worked?

$345 = 45 + 60H$	Substitute.
$345 - 45 = 60H$	Subtract 45 from both sides.
$300 = 60H$	
$H = 300 \div 60 = 5$	Divide both sides by 60.

ANSWER **H = 5 hours**

EXAMPLE 9.5

The length of a belt around two pulleys is $L = 2d + 3.26(r + R)$. With $r = 5$ in, $R = 9$ in and $L = 90$ in, find the distance d between centres to the nearest inch.

$90 = 2d + 3.26(5 + 9)$	Substitute.
$90 = 2d + 3.26(14)$	Simplify inside the parentheses.
$90 = 2d + 45.64$	Multiply.
$90 - 45.64 = 2d$	Subtract 45.64 from both sides.
$44.36 = 2d$	
$d = 22.18$	Divide both sides by 2.
$d = 22$	Round to the nearest inch.

ANSWER **$d = 22$ in**

WATCH OUT **Subtract first, then divide**

In $3200 = 2L + 900$, dividing 3200 by 2 first is wrong unless the 900 is divided too. Clear the added number first, and the mistake cannot happen.

When the coefficient is negative

Some formulas subtract a multiple of the unknown. The steps are the same; just carry the negative sign along.

EXAMPLE 9.6

Air temperature falls with altitude: $T = -0.002H + G$, where G is the ground temperature in °F and H is the height in feet. If $G = 70$°F, at what height is $T = 32$°F?

$32 = -0.002H + 70$	Substitute.
$32 - 70 = -0.002H$	Subtract 70 from both sides.
$-38 = -0.002H$	A smaller number minus a larger one is negative.
$H = \frac{-38}{-0.002} = 19{,}000$	Divide both sides by −0.002. A negative divided by a negative is positive.

ANSWER **$H = 19{,}000$ ft**

9.4 Equations with fractions

Sheet-metal and carpentry formulas are full of fractions, and the answers are expected as exact fractions. The steps do not change. Only the arithmetic does, so take it slowly.

EXAMPLE 9.7

The allowance for a Pittsburgh lock seam is $A = 2w + \frac{3}{16}$ in, where w is the width of the pocket. If $A = \frac{7}{8}$ in, find w as a fraction in lowest terms.

$\frac{7}{8} = 2w + \frac{3}{16}$	Substitute.
$\frac{7}{8} - \frac{3}{16} = 2w$	Subtract $\frac{3}{16}$ from both sides.
$\frac{14}{16} - \frac{3}{16} = \frac{11}{16} = 2w$	Common denominator 16: $\frac{7}{8} = \frac{14}{16}$.
$w = \frac{11}{16} \div 2 = \frac{11}{32}$	Divide both sides by 2. Dividing a fraction by 2 **doubles its denominator**.
Check: $2(\frac{11}{32}) + \frac{3}{16} = \frac{11}{16} + \frac{3}{16} = \frac{14}{16} = \frac{7}{8}$ ✓	

ANSWER $w = \frac{11}{32}$ **in**

WATCH OUT **Two traps in the lock-seam problem**

Adding instead of subtracting. The $\frac{3}{16}$ is added to 2w, so it must be *subtracted* from both sides. Adding gives $\frac{17}{16}$, and the final answer comes out far too large.

Multiplying instead of dividing. 2w means 2 times w, so undo it by dividing by 2. Half of $\frac{11}{16}$ is $\frac{11}{32}$, a smaller piece. An answer bigger than A itself cannot be right: the pocket is only part of the allowance.

CALCULATOR TIP **Checking with decimals**

Even when the answer must be a fraction, a decimal check takes seconds: 0.875 − 0.1875 = 0.6875, and 0.6875 ÷ 2 = 0.34375. Then 11 ÷ 32 = 0.34375. They agree.

PRACTICE 9.2 **Two-step equations**

Substitute and solve. Show every substitution and the steps.

1. P = 2L + 2W. P = 2,600 ft, W = 350 ft. Find L.
2. P = 2L + 2W. P = 94 in, L = 28.5 in. Find W.

3. **Office services.** $T = 45 + 60H$. $T = \$435$. Find H.
4. **Water/wastewater.** $P = 0.52D + 14.7$. $P = 30.3$ psi. Find D.
5. **Electronics.** $R = 0.02T + 5$. $R = 5.36$. Find T.
6. **Meteorology.** $T = -0.002H + G$. $G = 80°F$, $T = 44°F$. Find H.
7. **Sports.** Maximum heart rate $R = -0.8A + 176$. $R = 140$. Find the age A.
8. **Automotive.** $P + 2T = C$. $P = 3.250$ in, $C = 3.625$ in. Find T (do not round).
9. **Trades management.** $A = RT + M$. $A = \$2{,}900$, $M = \$740$, $R = \$72$ per hour. Find T.
10. **Machine trades.** $L = 2d + 3.26(r + R)$. $r = 4$ in, $R = 6$ in, $L = 70$ in. Find d to the nearest inch.
11. **Sheet metal.** $A = 2w + \frac{3}{16}$. $A = \frac{11}{16}$ in. Find w in lowest terms.
12. **Sheet metal.** $A = 2w + \frac{3}{16}$. $A = \frac{15}{16}$ in. Find w in lowest terms.
13. **Sheet metal.** $A = 2w + \frac{3}{16}$. $A = 1\frac{1}{8}$ in. Find w in lowest terms.
14. $3x + 8 = 29$
15. $5y - 12 = 48$
16. $\frac{n}{4} + 6 = 15$

CHAPTER SUMMARY

- An equation stays true when the same thing is done to both sides.
- Solve by undoing operations with their inverses until the unknown stands alone.
- Substitute first, simplify each side, then solve.
- In a two-step equation undo addition or subtraction first, then multiplication or division.
- With fractions, find a common denominator to subtract; dividing a fraction by 2 doubles its denominator.
- Always check the answer in the original formula.

CHAPTER 9 REVIEW

1. $x + 9.5 = 21$
2. $7m = 94.5$
3. $\frac{k}{12} = 2.5$
4. $4x + 7 = 39$
5. $6y - 15 = 57$
6. $D = RT$. $D = 1{,}050$ mi, $R = 420$ mph. Find T.
7. $I = \frac{E}{R}$. $I = 2.4$ A, $R = 50\ \Omega$. Find E.
8. $V = LWH$. $V = 4{,}200$ cu ft, $L = 48$ ft, $W = 8.5$ ft. Find H to the nearest tenth.
9. $P = 2L + 2W$. $P = 4{,}100$ ft, $W = 600$ ft. Find L.
10. $T = 45 + 60H$. $T = \$255$. Find H.
11. $P = 0.52D + 14.7$. $P = 40.7$ psi. Find D.

12. $R = -0.8A + 176$. $R = 152$. Find A.

13. $A = RT + M$. $A = \$4{,}150$, $M = \$1{,}030$, $R = \$78$. Find T.

14. $L = 2d + 3.26(r + R)$. $r = 7$ in, $R = 7$ in, $L = 100$ in. Find d to the nearest inch.

15. $A = 2w + \frac{3}{16}$. $A = \frac{13}{16}$ in. Find w in lowest terms.

16. $A = 2w + \frac{3}{16}$. $A = \frac{3}{4}$ in. Find w in lowest terms.

17. $S = \frac{\pi dn}{12}$. $S = 5{,}000$ ft/min, $d = 6$ in, $\pi = 3.14$. Find n to the nearest hundred rpm.

18. A student solves $\frac{5}{8} = 2w + \frac{1}{8}$ and gets $w = 1\frac{1}{2}$. Find the correct answer and name the likely mistakes.

APPLIED TECHNICAL MATH

Part 4 · Geometry

How much trim goes around it? How much material covers it? Part 4 answers those two questions, perimeter and area, for every flat shape met in the trades.

CHAPTER 10

Perimeter and Area of Quadrilaterals

Baseboard is bought by the foot and flooring by the square foot, for the same room. One measures the distance around; the other measures the surface inside. Mixing them up is the costliest mistake in estimating.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Tell perimeter from area and use the correct unit for each.
- Find the perimeter and area of rectangles and squares.
- Find the perimeter and area of parallelograms and trapezoids.
- Identify the height of a figure and tell it from a slanted side.
- Work backward from an area or perimeter to a missing dimension.

KEY TERMS

perimeter	The distance around a flat figure. Measured in length units: in, ft, m.
area	The amount of surface inside a flat figure. Measured in square units: sq in, sq ft, m^2.
quadrilateral	A figure with four straight sides.
parallel	Lines that run the same direction and never meet.
base	The side a figure is thought of as resting on.
height	The distance straight up from the base, measured at a right angle to it. Also called altitude.

10.1 Perimeter and area

Perimeter is a length: walk around the edge of the figure and measure how far you went. **Area** is a count of squares: how many 1-by-1 squares it takes to cover the figure.

	Perimeter	Area
Measures	distance around the edge	surface inside the edge
Units	in, ft, yd, m, cm	sq in, sq ft, sq yd, m^2, cm^2
Found by	**adding** the sides	**multiplying** two dimensions
Used for	trim, fence, gasket, weld length	paint, flooring, sheet metal, insulation

KEY RULE **The perimeter of any straight-sided figure**

Add the lengths of all its sides.

The special perimeter formulas are only shortcuts for this.

WATCH OUT **Units tell you which one you have**

If two lengths in feet are multiplied, the answer is in **square feet**. If they are added, the answer is in **feet**. An area given in plain feet, or a perimeter given in square feet, is wrong before the number is even checked.

All dimensions must be in the **same unit** before they are added or multiplied. A panel 4 ft by 18 in is 4 ft by 1.5 ft, or 48 in by 18 in.

10.2 Rectangles and squares

A **rectangle** has four right angles. Its opposite sides are equal, so it has two lengths and two widths. A **square** is a rectangle with all four sides equal.

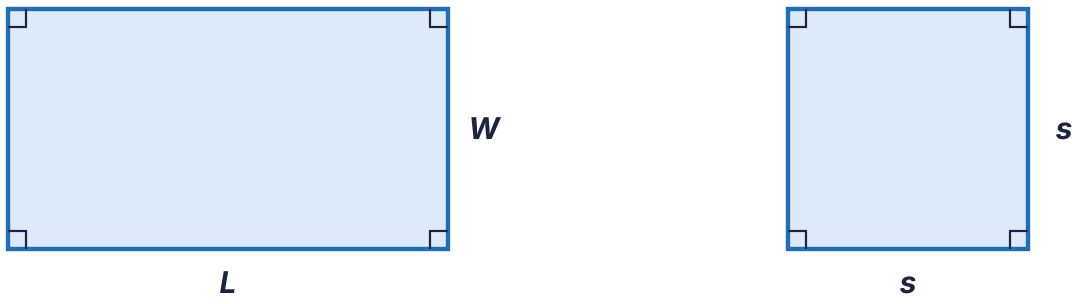


Figure 10.1 A rectangle with length L and width W, and a square with side s.

KEY RULE **Rectangle**

Perimeter: $P = 2L + 2W$

Area: $A = LW$

KEY RULE **Square**

Perimeter: $P = 4s$

Area: $A = s^2$

EXAMPLE 10.1

A rectangular return-air grille opening is 30 in long and 14 in wide. Find its perimeter and its area.

$P = 2L + 2W$	Perimeter formula.
$P = 2(30) + 2(14)$	Substitute.
$P = 60 + 28 = 88$	Multiply, then add.
$A = LW$	Area formula.
$A = (30)(14) = 420$	Substitute and multiply.

ANSWER **P = 88 in; A = 420 sq in**

EXAMPLE 10.2

A square concrete pad is 7.5 ft on a side. Find its perimeter and area.

$P = 4s = 4(7.5) = 30$	Four equal sides.
$A = s^2 = (7.5)^2 = 56.25$	7.5 × 7.5. Do not round unless told to.

ANSWER **P = 30 ft; A = 56.25 sq ft**

Working backward

If the area or perimeter is known and one dimension is missing, substitute what is known and solve, exactly as in Chapter 9.

EXAMPLE 10.3

A rectangular duct must have a cross-section of 216 sq in. It can be only 9 in deep. How wide must it be?

$A = LW$	Formula.
$216 = L(9)$	Substitute.
$L = 216 \div 9 = 24$	Divide both sides by 9.

ANSWER **24 in wide**

EXAMPLE 10.4

A square room has an area of 196 sq ft. How long is each wall?

$A = s^2$	Formula.
$196 = s^2$	Substitute.
$s = \sqrt{196} = 14$	The inverse of squaring is the square root. Take the root of both sides.

ANSWER **14 ft**

10.3 Parallelograms

A **parallelogram** has two pairs of parallel sides. It looks like a rectangle pushed over. Its sides are a base b and a slanted side a.

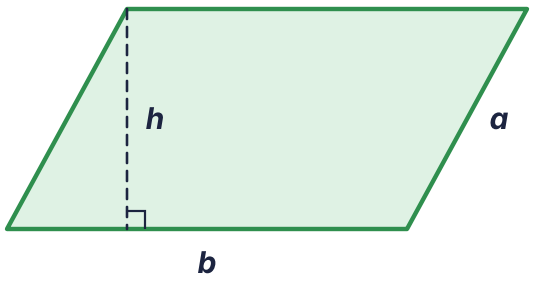


Figure 10.2 A parallelogram. The height h is perpendicular to the base; the side a is slanted.

KEY RULE **Parallelogram**

Perimeter: $P = 2a + 2b$

Area: $A = bh$

Why base times **height** and not base times side? Cut the triangle off one end of a parallelogram and slide it to the other end. The result is a rectangle with the same base and the same height. The slanted side does not appear in the area at all.

WATCH OUT **Height is never the slanted side**

The height is the straight-up distance between the two parallel sides. It is usually drawn as a dashed line with a small square where it meets the base. Use the slanted side for **perimeter** and the height for **area**.

EXAMPLE 10.5

A parallelogram has a base of 18 in, a slanted side of 10 in and a height of 8 in. Find its perimeter and area.

$P = 2a + 2b = 2(10) + 2(18)$	The perimeter uses the two side lengths.
$P = 20 + 36 = 56$	
$A = bh = (18)(8) = 144$	The area uses the base and the height. The 10 is not used.

ANSWER **P = 56 in; A = 144 sq in**

10.4 Trapezoids

A **trapezoid** has exactly one pair of parallel sides. Those two sides are its bases, b_1 and b_2. They are different lengths. The height is the perpendicular distance between them.

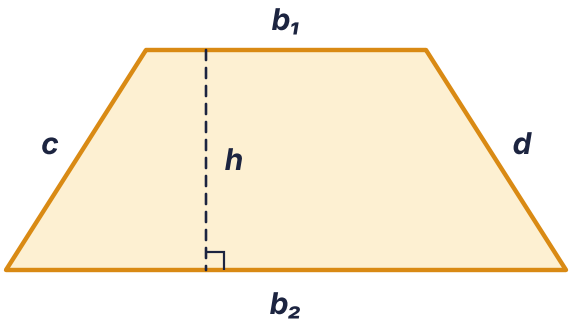


Figure 10.3 A trapezoid with bases b_1 and b_2, height h and slanted sides c and d.

KEY RULE **Trapezoid**

Perimeter: $P = b_1 + b_2 + c + d$ (the sum of the four sides)

Area: $A = \frac{(b_1 + b_2)h}{2}$

The area formula says: average the two bases, then multiply by the height. Two identical trapezoids, one flipped, fit together into a parallelogram whose base is $b_1 + b_2$. One trapezoid is half of that.

EXAMPLE 10.6

A trapezoidal transition panel has parallel sides of 12 in and 20 in, a height of 9 in, and slanted sides of 9.8 in each. Find the perimeter and the area.

$P = 12 + 20 + 9.8 + 9.8 = 51.6$	Add all four sides.
$A = \frac{(b_1 + b_2)h}{2}$	Area formula.
$A = \frac{(12 + 20)(9)}{2}$	Substitute.
$A = \frac{(32)(9)}{2} = \frac{288}{2}$	Parentheses first, then multiply across the top.
$A = 144$	Divide by 2.

ANSWER **P = 51.6 in; A = 144 sq in**

WATCH OUT **Add the bases before multiplying**

$(12 + 20)(9)$ is $32 \times 9 = 288$. Typing $12 + 20 \times 9$ gives 192, because the calculator multiplies first. Keep the parentheses.

EXAMPLE 10.7 Working backward

A trapezoid has an area of 150 sq ft. Its bases are 10 ft and 15 ft. Find its height.

$150 = \frac{(10 + 15)h}{2}$	Substitute.
$150 = \frac{25h}{2}$	Simplify.
$300 = 25h$	Multiply both sides by 2 to clear the fraction.
$h = 12$	Divide both sides by 25.

ANSWER **12 ft**

ON THE JOB **Estimating materials**

Area tells how much to buy; unit price tells what it costs. A 12 ft by 15 ft room is 180 sq ft, or $180 \div 9 = 20$ sq yd. Carpet at \$32 per square yard costs $20 \times 32 = \$640$. Convert the area to the unit the material is sold in before pricing.

PRACTICE 10.1 Quadrilaterals

Show the formula and the substitution. Do not round.

1. Rectangle: L = 16 ft, W = 9 ft. Find P and A.
2. Rectangle: L = 8.5 in, W = 11 in. Find P and A.
3. Square: s = 13 cm. Find P and A.
4. Square: s = 2.4 m. Find P and A.
5. Parallelogram: b = 22 in, a = 9 in, h = 7 in. Find P and A.
6. Parallelogram: b = 6.5 ft, a = 4 ft, h = 3.2 ft. Find P and A.
7. Trapezoid: b_1 = 8 in, b_2 = 14 in, h = 6 in, slanted sides 6.7 in each. Find P and A.
8. Trapezoid: b_1 = 5 m, b_2 = 9 m, h = 4.5 m. Find A.
9. Trapezoid: b_1 = 3.5 ft, b_2 = 6 ft, h = 2 ft. Find A.
10. A rectangle has A = 322 sq in and W = 14 in. Find L.
11. A rectangle has P = 76 ft and L = 23 ft. Find W.
12. A square has A = 529 sq cm. Find s.
13. A square has P = 58 in. Find s and A.
14. A parallelogram has A = 195 sq in and b = 15 in. Find h.
15. A trapezoid has A = 84 sq ft, b_1 = 9 ft and b_2 = 12 ft. Find h.
16. A room is 14 ft by 18 ft. How many square yards of carpet does it need?
17. A wall 24 ft long and 8 ft high has a door opening 3 ft by 7 ft. How many square feet are to be painted?
18. How many feet of baseboard go around a 12 ft by 15.5 ft room, ignoring doors?

CHAPTER SUMMARY

- Perimeter is the distance around (add; length units). Area is the surface inside (multiply; square units).
- Rectangle: P = 2L + 2W, A = LW. Square: P = 4s, A = s^2.
- Parallelogram: P = 2a + 2b, A = bh. Trapezoid: P = sum of the sides, A = $(b_1 + b_2)h \div 2$.
- The height is perpendicular to the base and is never a slanted side.
- To find a missing dimension, substitute what is known and solve.

CHAPTER 10 REVIEW

1. Would you use perimeter or area to order (a) weather-stripping, (b) roofing felt?
2. Rectangle: L = 27 in, W = 16 in. Find P.
3. Rectangle: L = 27 in, W = 16 in. Find A.

4. Square: s = 9.5 ft. Find P.
5. Square: s = 9.5 ft. Find A.
6. Parallelogram: b = 30 cm, a = 12 cm, h = 10 cm. Find P.
7. Parallelogram: b = 30 cm, a = 12 cm, h = 10 cm. Find A.
8. Trapezoid: bases 16 in and 24 in, height 11 in. Find A.
9. Trapezoid: sides 16 in, 24 in, 11.7 in and 11.7 in. Find P.
10. A rectangle has A = 247 sq ft and L = 19 ft. Find W.
11. A square has A = 2.25 sq m. Find s.
12. A rectangle has P = 130 in and W = 22 in. Find L.
13. A trapezoid has A = 135 sq in and bases 12 in and 18 in. Find h.
14. A rectangular sheet is 4 ft by 30 in. Find its area in square feet.
15. A 20 ft by 24 ft floor is tiled at $4.50 per square foot. Find the cost.
16. Why does the slanted side of a parallelogram not appear in its area formula?

CHAPTER 11

Triangles

Roof trusses, gussets, braces and offsets are all triangles. The triangle is the one shape that cannot be pushed out of square, which is why it holds up so much of what is built.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Classify a triangle as scalene, isosceles, equilateral or right.
- Find the perimeter of any triangle and its area from base and height.
- Find the area of a triangle from its three sides with Hero's formula.
- Use the shortcut area formula for an equilateral triangle.
- Use the Pythagorean theorem to find a missing side of a right triangle.

KEY TERMS

scalene	A triangle with no equal sides.
isosceles	A triangle with two equal sides (and two equal angles).
equilateral	A triangle with three equal sides. Every angle is 60°.
right triangle	A triangle with one 90° angle.
hypotenuse	The longest side of a right triangle, opposite the right angle.
leg	Either of the two sides that form the right angle.
semi-perimeter	Half the perimeter.

11.1 Kinds of triangles

Triangles are named by their sides. The angles of any triangle add up to 180°.

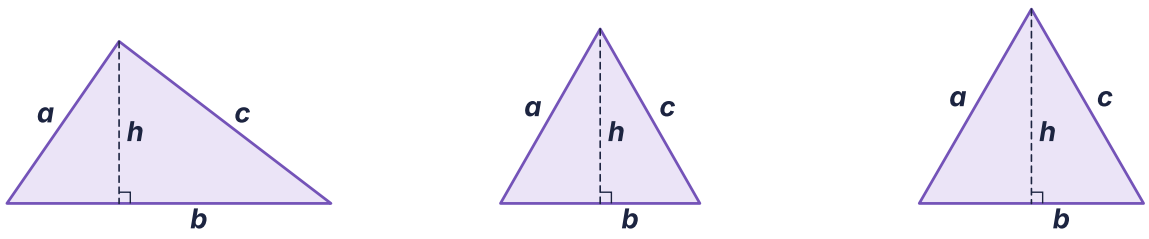


Figure 11.1 Scalene (no sides equal), isosceles (two sides equal), equilateral (all sides equal). The dashed line is the height.

KEY RULE **Any triangle**

Perimeter: $P = a + b + c$

Area: $A = \frac{bh}{2}$

where b is the base and h is the height measured at a right angle to that base.

A triangle is half of a parallelogram with the same base and height. That is where the “divide by 2” comes from.

EXAMPLE 11.1

A triangular gusset has a base of 16 in and a height of 9 in. Its other two sides are 15 in and 9.85 in. Find its area and perimeter.

$A = \frac{bh}{2} = \frac{(16)(9)}{2}$	Substitute base and height.
$A = \frac{144}{2} = 72$	Multiply, then halve.
$P = 16 + 15 + 9.85 = 40.85$	Perimeter is the sum of the three sides. The height is not a side.

ANSWER **A = 72 sq in; P = 40.85 in**

WATCH OUT **Do not forget the 2**

The most common triangle error is stopping at base × height. That is the area of the whole parallelogram, twice the triangle.

11.2 Hero’s formula: area from three sides

On a job the three sides of a triangle can be measured with a tape, but the height usually cannot. **Hero’s formula** (also written Heron’s) gives the area from the sides alone. It works for every triangle.

KEY RULE **Hero’s formula**

First find the semi-perimeter: $s = \frac{a + b + c}{2}$

Then: $A = \sqrt{(s(s - a)(s - b)(s - c))}$

EXAMPLE 11.2

Find the area of a triangle with sides 15 m, 20 m and 25 m. Round to the nearest tenth if needed.

$s = \frac{15 + 20 + 25}{2} = \frac{60}{2} = 30$	Semi-perimeter: add the sides and halve.
$A = \sqrt{(30(30 - 15)(30 - 20)(30 - 25))}$	Substitute s and each side.
$A = \sqrt{(30(15)(10)(5))}$	Do the three subtractions.
$A = \sqrt{22{,}500}$	Multiply the four numbers.
$A = 150$	Take the square root.

ANSWER **150 m²**

EXAMPLE 11.3

A triangular lot has sides of 70 ft, 90 ft and 110 ft. Find its area to the nearest square foot.

$s = \frac{70 + 90 + 110}{2} = 135$	Semi-perimeter.
$A = \sqrt{(135(65)(45)(25))}$	135 − 70 = 65, 135 − 90 = 45, 135 − 110 = 25.
$A = \sqrt{9{,}871{,}875}$	Multiply.
$A = 3141.9\ldots$	Square root.
$A = 3142$	Round.

ANSWER **3142 sq ft**

CALCULATOR TIP **Keying Hero's formula**

Find s first and write it down. Then enter `√( s × (s − a) × (s − b) × (s − c) )` with the outer parentheses, so the root covers the whole product. A negative number under the root means a subtraction was entered backward or the three sides cannot form a triangle.

11.3 The equilateral triangle

Because all three sides of an equilateral triangle are the same length *s*, its area depends on that one number.

KEY RULE **Equilateral triangle**

Perimeter: $P = 3s$

Area: $A = 0.433s^2$

EXAMPLE 11.4

Find the area of an equilateral triangle 8 in on a side.

$A = 0.433s^2$	Formula.
$A = 0.433(8)^2$	Substitute.
$A = 0.433(64)$	Exponent first.
$A = 27.712$	Multiply.

ANSWER **27.712 sq in**

The 0.433 is a rounded constant (it is $\sqrt{3} \div 4$), so this shortcut and Hero's formula agree to about three digits. Here s means the side. In Hero's formula s means the semi-perimeter. Check which formula you are in.

11.4 Right triangles and the Pythagorean theorem

A **right triangle** has one square corner. The two sides that form the corner are the **legs**, a and b. The side across from the right angle is the **hypotenuse**, c, and it is always the longest side.

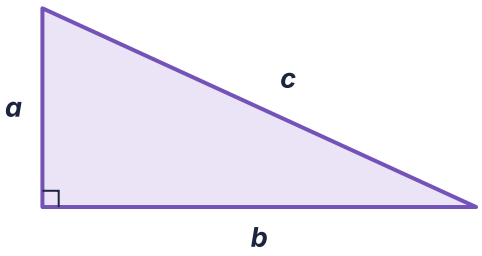


Figure 11.2 A right triangle. The legs a and b form the right angle; the hypotenuse c is opposite it.

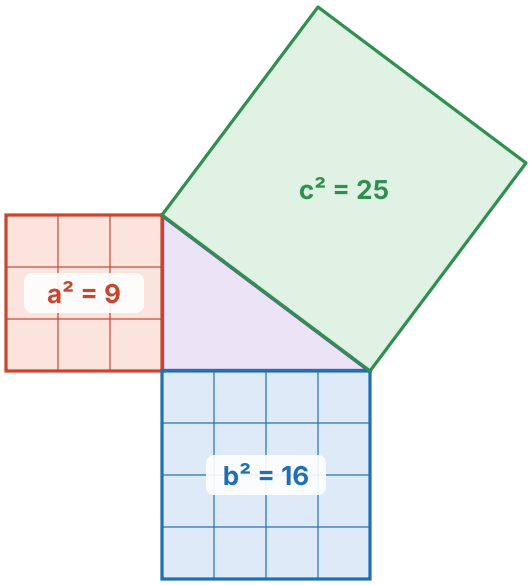


Figure 11.3 The squares built on the two legs of a 3-4-5 triangle together equal the square built on the hypotenuse.

KEY RULE **The Pythagorean theorem**

$a^2 + b^2 = c^2$

To find the hypotenuse: $c = \sqrt{(a^2 + b^2)}$

To find a leg: $a = \sqrt{(c^2 - b^2)}$

It is true **only** for right triangles.

EXAMPLE 11.5

A ladder's foot is 5 ft from a wall and its top touches the wall 12 ft up. How long is the ladder?

$c = \sqrt{(a^2 + b^2)}$	The ladder is the hypotenuse.
$c = \sqrt{(5^2 + 12^2)}$	Substitute the legs.
$c = \sqrt{(25 + 144)} = \sqrt{169}$	Square, then add.
$c = 13$	Square root.

ANSWER **13 ft**

EXAMPLE 11.6 Finding a leg

A 20-ft brace is fastened to a post at a point 16 ft above the ground. How far from the foot of the post does the brace meet the ground?

$a = \sqrt{(c^2 - b^2)}$	The brace is the hypotenuse; a leg is missing.
$a = \sqrt{(20^2 - 16^2)}$	Substitute. The hypotenuse goes first.
$a = \sqrt{(400 - 256)} = \sqrt{144}$	Square, then **subtract**.
$a = 12$	Square root.

ANSWER **12 ft**

WATCH OUT **Add for the hypotenuse, subtract for a leg**

The hypotenuse is the longest side, so finding it means adding the squares. A leg is shorter than the hypotenuse, so finding it means subtracting. If a "leg" comes out longer than the hypotenuse, the wrong operation was used.

And $\sqrt{(a^2 + b^2)}$ is **not** $a + b$. The squares must be added before the root is taken.

In a right triangle the legs are perpendicular, so one leg is the base and the other is the height: $A = \frac{ab}{2}$.

ON THE JOB **Squaring a corner with 3-4-5**

Builders check a corner by measuring 3 ft along one wall and 4 ft along the other. If the diagonal between the marks is exactly 5 ft, the corner is square, because $3^2 + 4^2 = 5^2$. Any multiple works: 6-8-10, 9-12-15, 12-16-20.

PRACTICE 11.1 **Triangles**

Show the formula and the substitution. Round only where the problem says.

1. Triangle: b = 14 in, h = 9 in. Find A.
2. Triangle: b = 7.5 ft, h = 4 ft. Find A.
3. Triangle with sides 9 cm, 12 cm and 17 cm. Find P.
4. Hero's formula: sides 6, 8, 10 in. Find A.
5. Hero's formula: sides 13, 14, 15 ft. Find A.
6. Hero's formula: sides 7, 10, 12 m. Find A to the nearest tenth.
7. Hero's formula: sides 20, 30, 40 ft. Find A to the nearest tenth.
8. Equilateral triangle: s = 10 in. Find P and A.

9. Equilateral triangle: s = 6.5 cm. Find A to the nearest tenth.
10. Right triangle: legs 9 in and 12 in. Find c.
11. Right triangle: legs 8 ft and 15 ft. Find c.
12. Right triangle: legs 7 m and 9 m. Find c to the nearest tenth.
13. Right triangle: c = 25 in, one leg 7 in. Find the other leg.
14. Right triangle: c = 30 ft, one leg 18 ft. Find the other leg.
15. Right triangle: c = 14 cm, one leg 9 cm. Find the other leg to the nearest tenth.
16. Right triangle: legs 10 in and 24 in. Find A and P.
17. A rafter rises 6 ft over a horizontal run of 14 ft. Find the rafter length to the nearest tenth.
18. A rectangular gate is 4 ft by 7 ft. How long is its diagonal brace, to the nearest hundredth?

CHAPTER SUMMARY

- Any triangle: $P = a + b + c$ and $A = bh \div 2$, with h perpendicular to b.
- Hero's formula finds area from the three sides: $s = (a + b + c) \div 2$, $A = \sqrt{(s(s - a)(s - b)(s - c))}$.
- Equilateral triangle: $P = 3s$, $A = 0.433s^2$.
- Right triangle: $a^2 + b^2 = c^2$. Add the squares to find the hypotenuse; subtract to find a leg.
- The hypotenuse is always the longest side.

CHAPTER 11 REVIEW

1. A triangle has sides 5, 5 and 8 in. What kind of triangle is it?
2. Each angle of an equilateral triangle measures how many degrees?
3. Triangle: b = 18 ft, h = 11 ft. Find A.
4. Triangle with sides 14, 17 and 21 in. Find P.
5. Hero's formula: sides 9, 12, 15 m. Find A.
6. Hero's formula: sides 10, 17, 21 cm. Find A.
7. Hero's formula: sides 8, 11, 13 ft. Find A to the nearest tenth.
8. Equilateral triangle: s = 12 in. Find A.
9. Equilateral triangle: s = 4.2 m. Find P.
10. Right triangle: legs 12 ft and 16 ft. Find c.
11. Right triangle: legs 5 in and 9 in. Find c to the nearest hundredth.
12. Right triangle: c = 26 cm, leg 10 cm. Find the other leg.
13. Right triangle: c = 12 ft, leg 7 ft. Find the other leg to the nearest tenth.
14. A 16-ft ladder leans against a wall with its foot 4 ft from the wall. How high does it reach, to the nearest tenth?
15. Is a triangle with sides 10, 24 and 26 a right triangle? Show why.
16. A triangle has area 54 sq in and base 12 in. Find its height.

CHAPTER 12

Circles, Hexagons and Composite Figures

Pipe, duct, shafting, tanks and bolt heads: round and six-sided shapes are everywhere in mechanical work. And real parts are seldom one simple shape. The last skill in the book is taking a complicated figure apart into shapes you already know.

AFTER THIS CHAPTER YOU WILL BE ABLE TO

- Name the parts of a circle and convert between radius and diameter.
- Find the circumference and area of a circle using $\pi = 3.14$.
- Work backward from circumference or area to the radius or diameter.
- Find the perimeter, area and across-corner and across-flat distances of a regular hexagon.
- Find the area of a composite figure by adding or subtracting simple shapes.

KEY TERMS

radius	The distance from the centre of a circle to its edge.
diameter	The distance across a circle through its centre: two radii.
circumference	The distance around a circle; its perimeter.
π (pi)	The circumference of any circle divided by its diameter; about 3.14.
regular hexagon	A six-sided figure with all sides equal and all angles 120°.
composite figure	A figure made of two or more simple shapes.

12.1 The circle

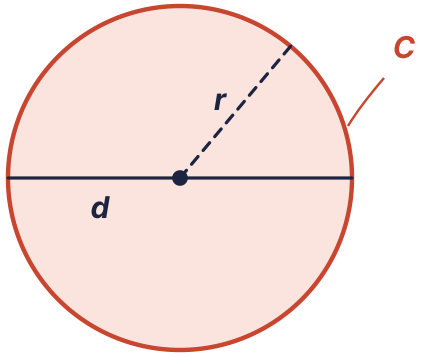


Figure 12.1 The radius r, the diameter d and the circumference C of a circle.

Measure around any circle, divide by the distance across, and the answer is always the same number, a little more than 3. That number is called **π** (pi). Its decimal never ends; this course uses **3.14**.

KEY RULE **Circle**

$d = 2r$ and $r = \frac{d}{2}$

Circumference: $C = \pi d$ or $C = 2\pi r$

Area: $A = \pi r^2$

Use $\pi = 3.14$.

EXAMPLE 12.1

A round duct has a diameter of 14 in. Find its circumference and its cross-sectional area.

$C = \pi d = (3.14)(14) = 43.96$	The circumference formula takes the diameter directly.
$r = \frac{d}{2} = \frac{14}{2} = 7$	The area formula needs the **radius**. Halve the diameter first.
$A = \pi r^2 = (3.14)(7)^2$	Substitute.
$A = (3.14)(49) = 153.86$	Square the radius first, then multiply by π.

ANSWER **C = 43.96 in; A = 153.86 sq in**

WATCH OUT **Three circle mistakes**

Using the diameter in $A = \pi r^2$. Halve it first. Using d makes the area four times too large.

Doubling instead of squaring. r^2 is r × r. For r = 7, r^2 is 49, not 14.

Squaring π. In πr^2 only the r is squared: 3.14 × 49, not $(3.14 \times 7)^2$.

Working backward

EXAMPLE 12.2

A tank measures 157 in around. What is its diameter?

$C = \pi d$	Formula.
$157 = 3.14d$	Substitute.
$d = 157 \div 3.14 = 50$	Divide both sides by 3.14.

ANSWER **50 in**

EXAMPLE 12.3

A circular opening must have an area of 200 sq in. Find the radius to the nearest tenth.

$A = \pi r^2$	Formula.
$200 = 3.14r^2$	Substitute.
$r^2 = 200 \div 3.14 = 63.694\ldots$	Divide both sides by 3.14.
$r = \sqrt{63.694\ldots} = 7.98\ldots$	Take the square root of both sides.
$r = 8.0$	Round to the nearest tenth.

ANSWER **8.0 in**

CALCULATOR TIP **The π key**

The calculator's π key uses 3.14159265…, which gives slightly different answers from 3.14 (153.94 instead of 153.86 above). When the course says to use 3.14, type 3.14.

12.2 The regular hexagon

A **regular hexagon** has six equal sides and six equal 120° angles. Bolt heads and nuts are hexagons. It can be measured two ways: **across the corners** (the distance *d* from one point to the opposite point) and **across the flats** (the distance *f* between opposite sides, which is the wrench size).

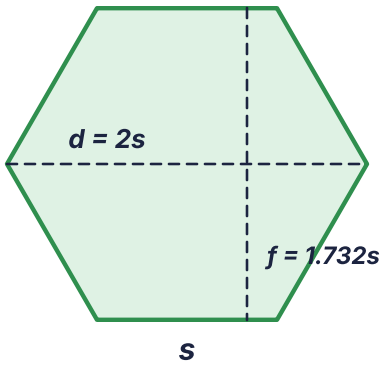


Figure 12.2 A regular hexagon with side s, distance across corners d and distance across flats f.

KEY RULE **Regular hexagon**

Perimeter: $P = 6s$

Area: $A = 2.6s^2$

Across corners: $d = 2s$

Across flats: $f = 1.732s$

EXAMPLE 12.4

A hexagonal bar has sides of 0.75 in. Find its perimeter, area, distance across corners and distance across flats.

$P = 6s = 6(0.75) = 4.5$	Six equal sides.
$A = 2.6s^2 = 2.6(0.75)^2 = 2.6(0.5625) = 1.4625$	Square the side first.
$d = 2s = 2(0.75) = 1.5$	Across corners.
$f = 1.732s = 1.732(0.75) = 1.299$	Across flats.

ANSWER **P = 4.5 in; A = 1.4625 sq in; d = 1.5 in; f = 1.299 in**

EXAMPLE 12.5 Working backward

A hexagonal patio is to cover 500 sq m. How long is each side, to the nearest tenth?

$A = 2.6s^2$	Formula.
$500 = 2.6s^2$	Substitute.
$s^2 = 500 \div 2.6 = 192.307\ldots$	Divide both sides by 2.6.
$s = \sqrt{192.307\ldots} = 13.867\ldots$	Square root of both sides.
$s = 13.9$	Round.

ANSWER **13.9 m**

EXAMPLE 12.6

A nut measures 1.5 in across the flats. How long is each side, to the nearest hundredth?

$f = 1.732s$	Formula.
$1.5 = 1.732s$	Substitute.
$s = 1.5 \div 1.732 = 0.866\ldots$	Divide.
$s = 0.87$	Round.

ANSWER **0.87 in**

PRACTICE 12.1 **Circles and hexagons**

Use $\pi = 3.14$. Do not round unless told to.

1. Circle: d = 10 in. Find C.
2. Circle: r = 6 ft. Find C.
3. Circle: r = 5 cm. Find A.
4. Circle: d = 18 in. Find A.
5. Circle: d = 3.5 m. Find C and A.
6. Circle: C = 94.2 in. Find d.
7. Circle: C = 50 ft. Find r to the nearest hundredth.
8. Circle: A = 314 sq in. Find r.
9. Circle: A = 120 sq ft. Find d to the nearest tenth.
10. Hexagon: s = 4 in. Find P and A.
11. Hexagon: s = 2.5 cm. Find d and f.
12. Hexagon: s = 12 ft. Find A.
13. Hexagon: A = 650 sq in. Find s to the nearest tenth.
14. Hexagon: d = 3 in across corners. Find s and f.
15. Hexagon: f = 2 in across flats. Find s to the nearest hundredth.
16. How far does a 26-in diameter wheel roll in one turn?

12.3 Composite figures

A **composite figure** is built from two or more simple shapes. No new formula is needed, only a plan.

KEY RULE **Area of a composite figure**

1. **Separate** the figure into shapes you know: rectangles, triangles, circles, trapezoids.
2. **Find** any dimension that is not given, by adding or subtracting the ones that are.
3. **Calculate** the area of each shape.
4. **Add** the areas of shapes that are joined. **Subtract** the areas of holes and cut-outs.

Adding parts

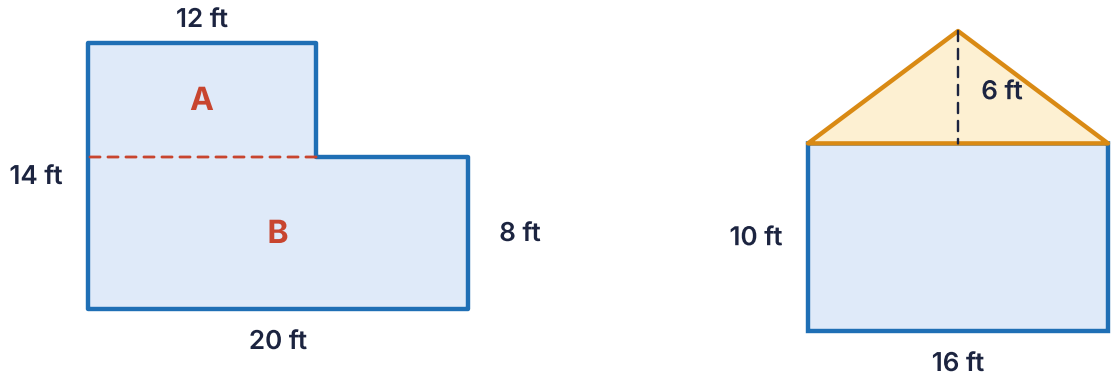


Figure 12.3 Left: an L-shaped floor split into rectangles A and B. Right: a gable wall, a rectangle with a triangle on top.

EXAMPLE 12.7

Find the area of the L-shaped floor in Figure 12.3.

A: 12 ft by 6 ft; B: 20 ft by 8 ft	Split along the dashed line. Rectangle A is 12 ft wide. Its height is 14 − 8 = 6 ft.
Area A = (12)(6) = 72	
Area B = (20)(8) = 160	
72 + 160 = 232	The parts are joined, so add.

ANSWER **232 sq ft**

The same floor can be done by subtraction: a full 20 by 14 rectangle is 280 sq ft, and the missing corner is 8 by 6 = 48 sq ft. 280 − 48 = 232 sq ft. Either plan must give the same answer, which makes a good check.

EXAMPLE 12.8

Find the area of the gable wall in Figure 12.3.

Rectangle: (16)(10) = 160	The lower part.
Triangle: $\frac{(16)(6)}{2} = 48$	The triangle's base is the width of the wall, 16 ft; its height is 6 ft.
160 + 48 = 208	Add.

ANSWER **208 sq ft**

Subtracting holes



Figure 12.4 Left: a plate with a rectangular opening. Right: a ring (a circle with a circular hole).

EXAMPLE 12.9

A 30 in by 20 in plate has an 18 in by 10 in opening cut from it. Find the area of metal that remains.

Outside: (30)(20) = 600	The whole plate.
Opening: (18)(10) = 180	The piece removed.
600 − 180 = 420	A hole is subtracted.

ANSWER **420 sq in**

EXAMPLE 12.10

A flat ring gasket has an outside radius of 8 ft and an inside radius of 5 ft. Find its area.

Outer circle: $(3.14)(8)^2 = 200.96$	
Inner circle: $(3.14)(5)^2 = 78.5$	
$200.96 - 78.5 = 122.46$	Subtract the hole.

ANSWER **122.46 sq ft**

WATCH OUT **Subtract areas, not radii**

The ring is **not** a circle of radius $8 - 5 = 3$. That would give 28.26 sq ft. Find each full area first, then subtract.

Half circles and quarter circles

A semicircle has half the area of the full circle: $A = \frac{\pi r^2}{2}$. Its curved edge is half the circumference, $\frac{\pi d}{2}$. When a semicircle sits on the end of a rectangle, its diameter equals the width of the rectangle.

EXAMPLE 12.11

A window is a rectangle 4 ft wide and 5 ft tall with a semicircle on top. Find its area.

Rectangle: $(4)(5) = 20$	
$r = 4 \div 2 = 2$	The semicircle's diameter is the 4-ft width.
Semicircle: $\frac{(3.14)(2)^2}{2} = \frac{12.56}{2} = 6.28$	Half of a full circle.
$20 + 6.28 = 26.28$	Add.

ANSWER **26.28 sq ft**

Perimeter of a composite figure

The perimeter of a composite figure is the distance around its **outside edge only**. Do not count the dashed lines where the parts were split; those lines are inside the figure. For the L-shaped floor: $20 + 8 + 8 + 6 + 12 + 14 = 68$ ft. For the window above: $4 + 5 + 5$ plus the half circumference $(3.14)(4) \div 2 = 6.28$, for 20.28 ft in all.

PRACTICE 12.2 **Composite figures**

Sketch each figure, split it into simple shapes, and show each area. Use $\pi = 3.14$.

1. An L-shaped room is 24 ft by 18 ft overall with a 10 ft by 6 ft corner missing. Find its area.

2. Find the perimeter of the room in problem 1.
3. A gable wall is 28 ft wide and 9 ft high to the eaves, with a triangular gable 7 ft high. Find its area.
4. A 36 in by 24 in panel has a 12 in by 12 in square opening. Find the remaining area.
5. A 20 in by 20 in plate has a circular hole 8 in in diameter. Find the remaining area.
6. A ring has an outside diameter of 12 in and an inside diameter of 6 in. Find its area.
7. A table top is a rectangle 6 ft by 3 ft with a semicircle on each 3-ft end. Find its area.
8. Find the perimeter of the table top in problem 7.
9. A wall 30 ft by 9 ft has two windows 4 ft by 5 ft and a door 3 ft by 7 ft. How much wall is left to paint?
10. A trapezoid with bases 10 in and 16 in and height 8 in has a rectangle 16 in by 5 in attached to its longer base. Find the total area.
11. A square 10 cm on a side has a quarter circle of radius 10 cm removed from one corner. Find the remaining area.
12. A hexagonal nut has sides of 0.5 in and a round hole 0.5 in in diameter. Find the area of its face to the nearest hundredth.

CHAPTER SUMMARY

- Circle: $d = 2r$, $C = \pi d$, $A = \pi r^2$, with $\pi = 3.14$. Use the radius for area.
- Regular hexagon: $P = 6s$, $A = 2.6s^2$, across corners $d = 2s$, across flats $f = 1.732s$.
- To work backward from an area with a squared term, divide first, then take the square root.
- Composite area: separate, find each area, add joined parts and subtract holes.
- Composite perimeter: follow the outside edge only.

CHAPTER 12 REVIEW

1. Circle: r = 9 in. Find d and C.
2. Circle: d = 22 ft. Find A.
3. Circle: C = 78.5 cm. Find d.
4. Circle: A = 50.24 sq in. Find r.
5. Circle: A = 90 sq ft. Find r to the nearest hundredth.
6. Hexagon: s = 7 in. Find P.
7. Hexagon: s = 7 in. Find A.
8. Hexagon: s = 7 in. Find d and f.
9. Hexagon: A = 400 sq m. Find s to the nearest tenth.
10. Hexagon: f = 3 in. Find s to the nearest hundredth.
11. An L-shaped slab is 30 ft by 20 ft overall with a 12 ft by 8 ft corner missing. Find its area.
12. A 48 in by 30 in sheet has two round holes, each 10 in in diameter. Find the remaining area.
13. A ring has an outside radius of 10 cm and an inside radius of 7 cm. Find its area.
14. A window is a 3 ft by 4 ft rectangle topped by a semicircle on the 3-ft side. Find its area.
15. A gable end is 24 ft wide, 8 ft to the eaves and 5 ft more to the peak. Find its area.

16. Why is the area of a ring with radii 10 and 7 not the area of a circle with radius 3?

Reference tables and formulas

Everything on these pages is developed in the chapters. They are collected here for quick use.

Standard units

Length	Weight	Liquid capacity
1 ft = 12 in	1 lb = 16 oz	1 pt = 2 cups
1 yd = 3 ft = 36 in	1 T = 2000 lb	1 qt = 2 pt
1 mi = 5280 ft = 1760 yd		1 gal = 4 qt

Area and volume

Area	Volume	Volume and capacity
1 sq ft = 144 sq in	1 cu ft = 1728 cu in	1 cu ft = 7.48 gal
1 sq yd = 9 sq ft	1 cu yd = 27 cu ft	1 gal = 231 cu in
1 sq yd = 1296 sq in	1 cu yd = 46,656 cu in	

Metric stairs (each step is one decimal place)

kilo	hecto	deka	unit	deci	centi	milli
1000	100	10	1	0.1	0.01	0.001
km	hm	dam	m	dm	cm	mm
kg	hg	dag	g	dg	cg	mg
kL	hL	daL	L	dL	cL	mL

Standard and metric

Length	Weight	Capacity
1 in = 2.54 cm	1 oz = 28.35 g	1 L = 1.06 qt
1 m = 3.28 ft	1 kg = 2.205 lb	1 gal = 3.785 L
1 mi = 1.61 km		

Every conversion in both directions

Category	Multiply by the factor	Divide by the factor	Factor
Length	ft → in	in → ft	12
Length	yd → ft	ft → yd	3
Length	mi → ft	ft → mi	5280
Weight	lb → oz	oz → lb	16
Weight	T → lb	lb → T	2000
Capacity	gal → qt; qt → pt; pt → cups	qt → gal; pt → qt; cups → pt	4; 2; 2
Area	sq ft → sq in	sq in → sq ft	144
Area	sq yd → sq ft	sq ft → sq yd	9
Area	sq yd → sq in	sq in → sq yd	1296
Volume	cu ft → cu in	cu in → cu ft	1728
Volume	cu yd → cu ft	cu ft → cu yd	27
Volume	cu ft → gal	gal → cu ft	7.48
Volume	gal → cu in	cu in → gal	231
Metric length	in → cm	cm → in	2.54
Metric length	m → ft	ft → m	3.28
Metric length	mi → km	km → mi	1.61
Metric weight	oz → g	g → oz	28.35
Metric weight	kg → lb	lb → kg	2.205
Metric capacity	L → qt	qt → L	1.06
Metric capacity	gal → L	L → gal	3.785

Percent

To find	Proportion	Shortcut
the part	$\frac{x}{\text{whole}} = \frac{\text{percent}}{100}$	whole × percent as a decimal
the whole	$\frac{\text{part}}{x} = \frac{\text{percent}}{100}$	part ÷ percent as a decimal
the percent	$\frac{\text{part}}{\text{whole}} = \frac{x}{100}$	part ÷ whole, point 2 places right

Geometry formulas (π = 3.14)

Figure	Perimeter	Area
Rectangle	$P = 2L + 2W$	$A = LW$
Square	$P = 4s$	$A = s^2$
Parallelogram	$P = 2a + 2b$	$A = bh$
Trapezoid	P = sum of the sides	$A = (b_1 + b_2)h \div 2$
Triangle	$P = a + b + c$	$A = bh \div 2$
Triangle from three sides (Hero)	$s = (a + b + c) \div 2$	$A = \sqrt{(s(s - a)(s - b)(s - c))}$
Equilateral triangle	$P = 3s$	$A = 0.433s^2$
Right triangle	$a^2 + b^2 = c^2$	$A = ab \div 2$
Regular hexagon	$P = 6s$; $d = 2s$; $f = 1.732s$	$A = 2.6s^2$
Circle	$C = \pi d = 2\pi r$; $d = 2r$	$A = \pi r^2$

Trade formulas used in this book

Trade	Formula	Meaning
Machine trades	$L = \pi dr \div 12$	linear speed of a wheel, ft/min (d in, r rpm)
Construction	$L = 25T^4 \div H^2$	crushing load of a square post, tons
Welding	$T = L \div A$	tensile strength, psi
Plumbing	$N = \sqrt{(D \div d)^5}$	small pipes equal to one large pipe
Manufacturing	$C = W^2DS \div 1.085$	feeder belt capacity, metric tons per hour
Hydraulics	$F = \pi PD^2 \div 4$	force on a piston, lb
Electronics	$C_T = C_1(C_2 + C_3) \div (C_1 + C_2 + C_3)$	series-parallel capacitance
Electronics	$L_T = L_1L_2 \div (L_1 + L_2)$	two inductors in parallel
Electronics	$L = R^2N^2 \div (9R + 10D)$	inductance of a coil, µH
Electrical	$I = E \div R$	Ohm's law
Automotive	$ihp = MEP \times D \times S \div 792{,}000$	indicated horsepower
Machine trades	$L = 2d + 3.26(r + R)$	belt length around two pulleys
Sheet metal	$A = 2w + \frac{3}{16}$	Pittsburgh lock allowance, in
Water/wastewater	$P = 0.52D + 14.7$	pressure at depth D ft, psi

Order of operations and rounding

Rule	Remember
Order of operations	1 grouping symbols; 2 exponents and roots; 3 × and ÷ left to right; 4 + and − left to right
Rounding	Find the rounding digit; look one digit right; 5 or more rounds up, 4 or less stays; zeros for whole places, drop decimal places
Solving	Substitute, simplify, undo + and − first, then × and ÷; do the same to both sides; check

Answers to practice sets and reviews

Every answer here was worked by the rules in its chapter. Rounded answers say how they were rounded in the question.

Chapter 1: Place Value and Rounding

Practice 1.1: Place value

1. 7
2. 2
3. 9
4. 4
5. 0
6. 0
7. 8
8. 6
9. 1
10. 3
11. 0.5
12. 0.0625
13. 1.3
14. 0.44
15. 2.05
16. 0.75

Practice 1.2: Rounding

1. 638.3
2. 638.27
3. 640
4. 600
5. 16,000
6. 15,862.095
7. 15,862
8. 15,862.1
9. 1.00
10. 1.0
11. 7,450
12. 7,400
13. 86.05
14. 250.0
15. 3,900
16. 71,400
17. 56
18. 0.044

Chapter review

1. 0
2. 5
3. 8
4. 0
5. 4
6. 3
7. 5,275.0
8. 5,300
9. 5,274.964
10. 5,275
11. 5,270
12. 5,274.96
13. 0.32
14. 1.068
15. 42.07
16. 0.375
17. 2,900 rpm
18. 7.47 µF
19. The student rounded twice (6.2481 → 6.25 → 6.3). The test digit is 4, so the answer is 6.2.
20. 20.00

Chapter 2: Fractions, Decimals and Percents

Practice 2.1: Fractions

1. $\frac{3}{4}$
2. $\frac{3}{4}$
3. $\frac{5}{16}$
4. $\frac{3}{8}$
5. $\frac{7}{8}$
6. $\frac{7}{8}$
7. $\frac{9}{16}$
8. $\frac{1}{4}$
9. $\frac{7}{4}$
10. $\frac{21}{8}$
11. $\frac{55}{16}$
12. $\frac{11}{2}$
13. $1\ \frac{3}{8}$
14. $2\ \frac{1}{4}$
15. $2\ \frac{5}{16}$

16. $1\frac{13}{32}$

Practice 2.2: Changing forms

1. $\frac{3}{8} = 0.375 = 37.5\%$
2. $\frac{4}{5} = 0.8 = 80\%$
3. $\frac{9}{20} = 0.45 = 45\%$
4. $\frac{7}{16} = 0.4375 = 43.75\%$
5. $\frac{13}{200} = 0.065 = 6.5\%$
6. $\frac{1}{8} = 0.125 = 12.5\%$
7. $\frac{9}{20} = 0.45 = 45\%$
8. $1\frac{1}{2} = 1.5 = 150\%$
9. $\frac{1}{50} = 0.02 = 2\%$
10. $\frac{13}{16} = 0.8125 = 81.25\%$
11. $\frac{5}{16} = 0.3125 = 31.25\%$
12. $\frac{7}{8} = 0.875 = 87.5\%$

Practice 2.3: Adding and subtracting

1. $\frac{5}{8}$
2. $\frac{13}{16}$
3. $\frac{11}{16}$
4. $\frac{7}{16}$
5. $3\frac{7}{8}$
6. $7\frac{1}{16}$
7. $3\frac{5}{8}$
8. $6\frac{9}{16}$
9. 21.25 in
10. 31.875 in
11. 31.625 in
12. 35.625 in

Chapter review

1. $\frac{9}{16}$
2. $\frac{5}{8}$
3. 0.875
4. 0.1875
5. 2.5625
6. $\frac{5}{8}$
7. $\frac{7}{20}$
8. $\frac{3}{16}$
9. 42%
10. 7.5%
11. 180%
12. 0.64
13. 0.07
14. 1.125
15. 0.005
16. 68.75%
17. $\frac{5}{8}$
18. $5\frac{3}{8}$ in ($0.40 \times 16 = 6.4 \rightarrow \frac{6}{16}$)
19. $6\frac{3}{16}$
20. $4\frac{3}{8}$
21. 17.9375 in
22. $1\frac{3}{16}$ in
23. $\frac{13}{16}$ (it is 0.8125)

Chapter 3: The Percent Proportion

Practice 3.1: The three types

1. 120
2. 7.5%
3. 500
4. 9
5. 150
6. 12.5%
7. 15.6
8. 125%
9. 600
10. 102
11. 0.625%
12. 45
13. 4.5
14. 27.5%
15. 200
16. 24

Practice 3.2: Applications

1. 34%
2. 1,056 W
3. $25.97
4. $1,160.00
5. 352 sq ft
6. 12% failed; 88% passed
7. $8,400
8. 80 gal
9. 5%
10. 15%
11. 446.5 Ω to 493.5 Ω
12. $650.00

Chapter review

1. 91
2. 120
3. 15%
4. 36
5. 125%
6. 1,500
7. 30
8. $\frac{24}{64} = \frac{x}{100}$
9. 63
10. 140
11. 85%
12. $480
13. 24%
14. 64%
15. $1,300
16. 8% was used as 0.8 instead of 0.08 (or the ÷ 100 was skipped). 8 × 50 ÷ 100 = 4.

Chapter 4: Standard Units of Measure

Practice 4.1: Length, weight and capacity

1. 108 in
2. 8.5 ft
3. 13.5 ft
4. 16.7 yd
5. 13,200 ft
6. 1.7 mi
7. 80 oz
8. 4.5 lb
9. 6,400 lb
10. 2.8 T
11. 24 qt
12. 4.5 gal
13. 3.5 qt
14. 16 pt
15. 5 yd
16. 1,320 yd

Practice 4.2: Area and volume

1. 5.9 sq ft
2. 112.5 sq ft
3. 2,592 sq in
4. 1.5 sq yd
5. 44.4 sq yd
6. 66.7 cu yd
7. 2.3 cu ft
8. 445.5 cu ft
9. 6,048 cu in
10. 59.8 gal
11. 8.7 gal
12. 5.3 cu ft
13. 693 cu in
14. 93,312 cu in
15. 16 cu ft
16. 648 sq in

Chapter review

1. 168 in
2. 16.7 ft
3. 112 oz
4. 3.5 T
5. 2.5 gal
6. 4.2 sq ft
7. 225 sq ft
8. 18.5 cu yd
9. 4,320 cu in
10. 89.8 gal
11. 4.3 gal
12. 10 cu ft
13. A square foot is 12 in by 12 in, so it holds 12 × 12 = 144 square inches.
14. 30 sq yd
15. 5 cu yd
16. 224.4 gal
17. 126,720 in
18. 12,705 cu in

Chapter 5: The Metric System

Practice 5.1: Metric to metric

1. 450 cm
2. 38 cm
3. 2,400 m
4. 0.75 kg
5. 600 mL
6. 1.25 L
7. 350 mm
8. 8.2 km
9. 4.6 dag
10. 320 L
11. 85 g
12. 0.95 m
13. 700 dm
14. 12.5 g
15. 400 mm
16. 0.64 L

Practice 5.2: Metric and standard

1. 30.5 cm
2. 31.5 in
3. 82 ft
4. 30.5 m
5. 42.2 km
6. 6.2 mi
7. 226.8 g
8. 17.6 oz
9. 154.4 lb
10. 99.8 kg
11. 21.2 qt
12. 56.8 L
13. inch
14. litre
15. ounce
16. mile

Practice 5.3: Conversions by category

1. 275 cm
2. 0.64 m
3. 3,400 m
4. 45.7 cm
5. 18.3 m
6. 88.6 km
7. 2,200 g
8. 0.35 g
9. 4,800 cg
10. 680.4 g
11. 35.3 oz
12. 99.2 lb
13. 1,500 mL
14. 2.4 L
15. 3.6 L
16. 28.3 L
17. 15.9 gal
18. 28.4 L

Chapter review

1. 6,800 mm
2. 4.25 m
3. 750 g
4. 0.9 L
5. 0.52 dag
6. 120,000 cm
7. 91.4 cm
8. 59.1 in
9. 11 lb
10. 72.6 kg
11. 42.4 qt
12. 18.9 L
13. 24.5 L
14. 248.4 mi
15. a metre (1 m = 3.28 ft, and a yard is 3 ft)
16. a cubic foot (it holds 7.48 gallons)
17. 78.7 in
18. 1,361 g
19. 0.75 in
20. 4 places to the right

Chapter 6: Rulers, Gauges and Tolerance

Practice 6.1: Reading a rule

1. 14 mm
2. 37 mm
3. 52 mm
4. 78 mm
5. 95 mm
6. $\frac{3}{16}$ in
7. $\frac{3}{4}$ in
8. $1\ \frac{3}{8}$ in
9. $1\ \frac{13}{16}$ in
10. $2\ \frac{1}{2}$ in
11. $2\ \frac{7}{8}$ in
12. $3\ \frac{9}{16}$ in
13. $3\ \frac{7}{8}$ in
14. 5.2 cm
15. 64 mm
16. $1\ \frac{3}{4}$ in

Practice 6.2: Reading gauges

1. 5 psi
2. about 65 psi
3. 40 °F
4. about 340 °F
5. 0.5 volts
6. about 13.5 volts
7. 20 psi
8. about 210 psi
9. 2 amps
10. about 47 amps
11. 25 kPa
12. about 140 kPa

Practice 6.3: Tolerance and missing lengths

1. 2.49 in to 2.51 in; yes
2. 0.37 in to 0.38 in; no
3. 1.248 in to 1.252 in; yes
4. 3.95 in to 4.05 in; no
5. 0.617 in to 0.623 in; no (0.625)
6. 0.436 in to 0.444 in; yes (0.4375)

7. 23.125 in = 23 $\frac{1}{8}$ in
8. 37.625 in = 37 $\frac{5}{8}$ in
9. 14 in = 14 in
10. 31.75 in = 31 $\frac{3}{4}$ in
11. 135 Ω to 165 Ω
12. 92 $\frac{1}{2}$ in to 92 $\frac{3}{4}$ in

Chapter review

1. 73 mm
2. 64 mm
3. 2 $\frac{3}{4}$ in
4. 5 $\frac{7}{8}$ in
5. $\frac{3}{4}$ (quarter marks are longer than eighth marks)
6. 5
7. 95
8. about 1750
9. about 690 kPa
10. 1.735 in to 1.765 in
11. No. The upper limit is 0.502 in.
12. Yes. 11/16 = 0.6875, and the limits are 0.685 to 0.695.
13. 17.375 in
14. 6.875 in

Chapter 7: Order of Operations

Practice 7.1: Order of operations

1. 8
2. 17
3. 12
4. 23
5. 32
6. 32
7. 12
8. 41
9. 10
10. 22
11. 4
12. 3
13. 45.83
14. 18
15. 0
16. 11
17. -95
18. 5.71

Chapter review

1. 1 grouping symbols; 2 exponents and roots; 3 multiplication and division, left to right; 4 addition and subtraction, left to right.
2. 13
3. 18
4. 57
5. 192
6. 12
7. 8
8. 13
9. 13
10. 4
11. 3
12. 56
13. 10
14. 15.625
15. 7.47
16. The student added 10 + 5 first. Left to right: 30 − 10 = 20, then 20 + 5 = 25.
17. (7 + 9) ÷ (2 × 4) = 2
18. 48. The exponent applies to the 4 only and is done before multiplying: 3 × 16.

Chapter 8: Evaluating Formulas

Practice 8.1: Substitute and evaluate

1. -115
2. 162
3. $\frac{1}{40}$
4. 3
5. 100.625
6. 61.5 ft
7. 69
8. 15
9. 70,000 psi
10. 300 W
11. 3,140 ft/min
12. 324 tons
13. 32 pipes
14. 32 pipes
15. 50 mH
16. 7.88 µF
17. 1,324 µH
18. 187.5 hp
19. 196 lb
20. 2,240 t/h

Chapter review

1. 31
2. 48
3. 1.25
4. -96
5. 144.9
6. 24
7. 71,400 psi
8. 5,495 ft/min
9. 400 tons
10. 16 pipes
11. 3,733 t/h
12. 376.8 lb
13. 7.2 µF
14. 48 mH
15. W, D, S and C are variables; 1.085 is a constant.
16. 2(30), which is 60.

Chapter 9: Solving Formulas

Practice 9.1: One-step equations

1. 5.6 h
2. 15 ft
3. 1.372 m
4. 24 psi
5. 25 windings
6. 120 V
7. 15,000 lb
8. 17 ft
9. 8 ft
10. 17.5
11. 15.75
12. 27

Practice 9.2: Two-step equations

1. 950 ft
2. 18.5 in
3. 6.5 h
4. 30 ft
5. 18
6. 18,000 ft
7. 45 years
8. 0.1875 in
9. 30 h
10. 19 in
11. $\frac{1}{4}$ in
12. $\frac{3}{8}$ in
13. $\frac{15}{32}$ in
14. 7
15. 12
16. 36

Chapter review

1. 11.5
2. 13.5
3. 30
4. 8
5. 12
6. 2.5 h
7. 120 V
8. 10.3 ft
9. 1,450 ft
10. 3.5 h
11. 50 ft
12. 30 years
13. 40 h
14. 27 in
15. $\frac{5}{16}$ in
16. $\frac{9}{32}$ in
17. 3,200 rpm
18. w = $\frac{1}{4}$. The student added 1/8 instead of subtracting it and multiplied by 2 instead of dividing.

Chapter 10: Perimeter and Area of Quadrilaterals

Practice 10.1: Quadrilaterals

1. P = 50 ft; A = 144 sq ft
2. P = 39 in; A = 93.5 sq in
3. P = 52 cm; A = 169 cm^2
4. P = 9.6 m; A = 5.76 m^2
5. P = 62 in; A = 154 sq in
6. P = 21 ft; A = 20.8 sq ft
7. P = 35.4 in; A = 66 sq in
8. A = 31.5 m^2
9. A = 9.5 sq ft
10. 23 in
11. 15 ft
12. 23 cm
13. s = 14.5 in; A = 210.25 sq in
14. 13 in
15. 8 ft
16. 28 sq yd
17. 171 sq ft
18. 55 ft

Chapter review

1. (a) perimeter (b) area
2. 86 in
3. 432 sq in
4. 38 ft
5. 90.25 sq ft
6. 84 cm
7. 300 cm^2
8. 220 sq in
9. 63.4 in
10. 13 ft
11. 1.5 m
12. 43 in
13. 9 in
14. 10 sq ft
15. $2,160
16. Sliding the end triangle across turns it into a rectangle with the same base and height; only the perpendicular height matters.

Chapter 11: Triangles

Practice 11.1: Triangles

1. 63 sq in
2. 15 sq ft
3. 38 cm
4. 24 sq in
5. 84 sq ft
6. 35 m^2
7. 290.5 sq ft
8. P = 30 in; A = 43.3 sq in
9. 18.3 cm^2
10. 15 in
11. 17 ft
12. 11.4 m
13. 24 in
14. 24 ft
15. 10.7 cm
16. A = 120 sq in; P = 60 in
17. 15.2 ft
18. 8.06 ft

Chapter review

1. isosceles
2. 60°
3. 99 sq ft
4. 52 in
5. 54 m^2
6. 84 cm^2
7. 43.8 sq ft
8. 62.352 sq in
9. 12.6 m
10. 20 ft
11. 10.3 in
12. 24 cm
13. 9.7 ft
14. 15.5 ft
15. Yes: $10^2 + 24^2 = 100 + 576 = 676 = 26^2$.
16. 9 in

Chapter 12: Circles, Hexagons and Composite Figures

Practice 12.1: Circles and hexagons

1. 31.4 in
2. 37.68 ft
3. 78.5 cm^2
4. 254.34 sq in
5. C = 10.99 m; A = 9.61625 m^2
6. 30 in
7. 7.96 ft
8. 10 in
9. 12.4 ft
10. P = 24 in; A = 41.6 sq in
11. d = 5 cm; f = 4.33 cm
12. 374.4 sq ft
13. 15.8 in
14. s = 1.5 in; f = 2.598 in
15. 1.15 in
16. 81.64 in

Practice 12.2: Composite figures

1. 372 sq ft
2. 84 ft
3. 350 sq ft
4. 720 sq in
5. 349.76 sq in
6. 84.78 sq in
7. 25.065 sq ft
8. 21.42 ft
9. 209 sq ft
10. 184 sq in
11. 21.5 cm^2
12. 0.45 sq in

Chapter review

1. d = 18 in; C = 56.52 in
2. 379.94 sq ft
3. 25 cm

4. 4 in

5. 5.35 ft

6. 42 in

7. 127.4 sq in

8. d = 14 in; f = 12.124 in

9. 12.4 m

10. 1.73 in

11. 504 sq ft

12. 1,283 sq in

13. 160.14 cm^2

14. 15.5325 sq ft

15. 252 sq ft

16. Areas depend on the square of the radius: $10^2 - 7^2 = 51$, but $3^2 = 9$. Subtract the areas, not the radii.